

Chapter 4

Production Processes – Planning and Control

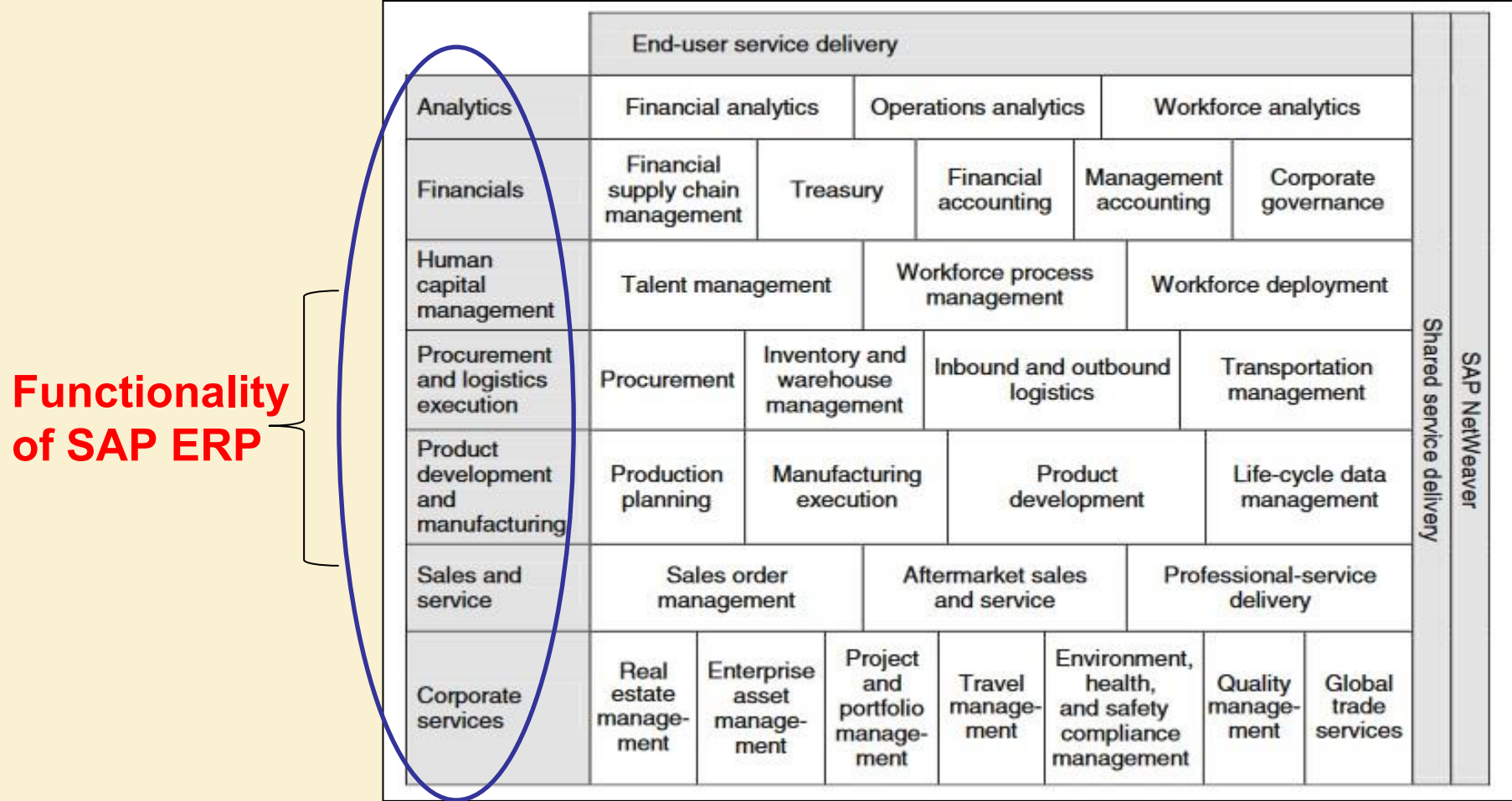
Learning objectives

- Production Development and Manufacturing.
- MRP - Materials Requirement Planning.
- MRP II – Manufacturing Resource Planning.
- Production Planning Master Data.
- Production Planning Processes.
- Manufacturing Execution Process.
- Production Planning and Execution in SAP S/4HANA (using Global Bike).

1. Production Development and Manufacturing

Solution Map of SAP ERP

- The SAP ERP solution map

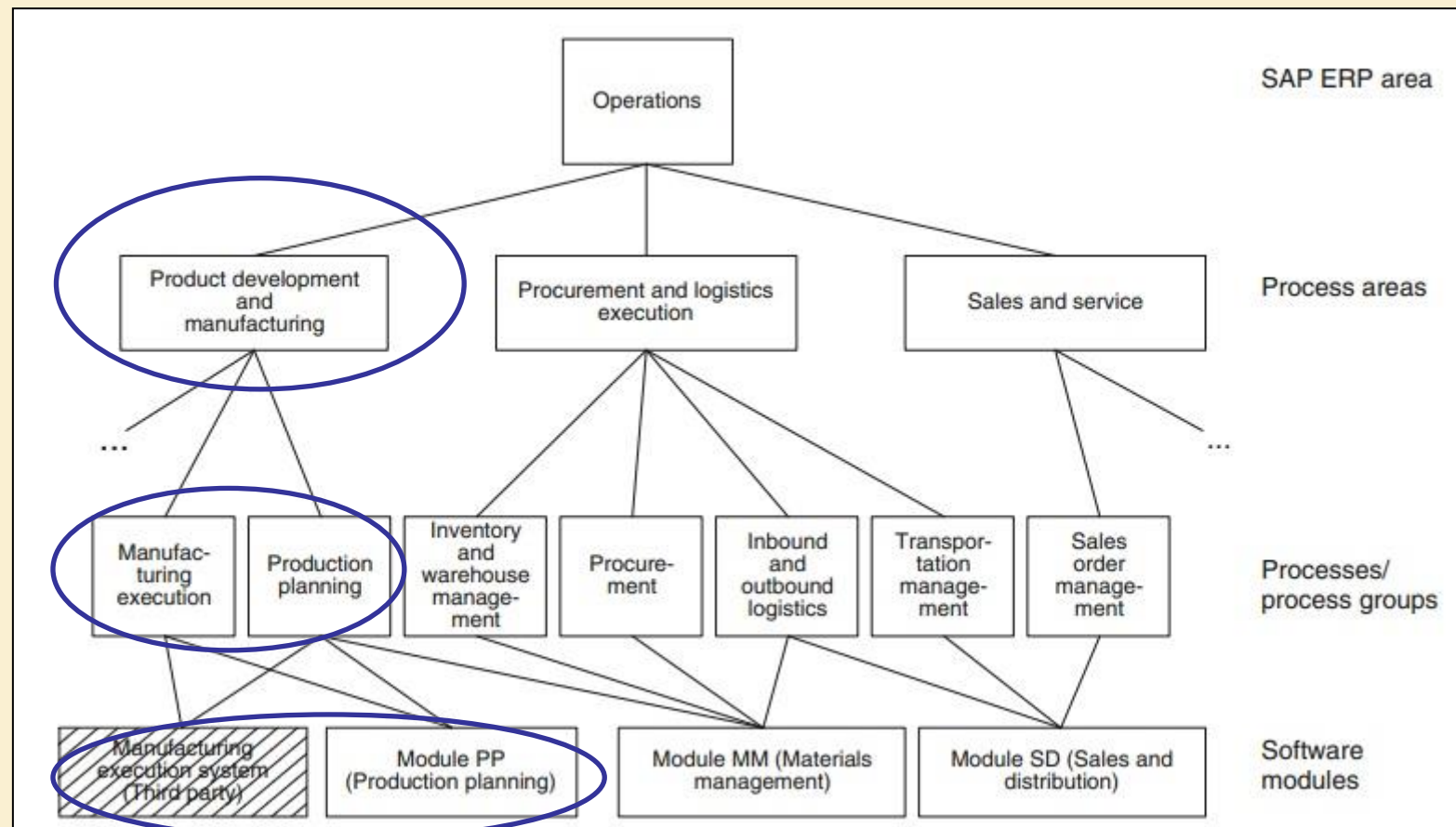


Production Development and Manufacturing

- The processes and tasks of product development and manufacturing are divided into the following subareas:
 - ❖ Production planning.
 - ❖ Manufacturing execution.
 - ❖ Product development.
 - ❖ Life-cycle data management.

Production Development and Manufacturing (Cont.)

- Relationship between solution map and SAP ERP modules



Production Development and Manufacturing

- Production planning

- Production planning includes determining dependent requirements, forward shifting, creating planned orders, lead-time scheduling, and capacity planning (long-, medium-, and short-term planning).
- Production planning more or less covers the MRP II functionality, that is, those parts of MRP II that are not supported by the procurement, inventory, and warehouse management parts.

Production Development and Manufacturing

- Production planning (Cont.)

- The functionality is related to material requirement planning. In particular, consumption- and requirements-driven material planning, comprising net requirements calculation, lot sizing, forward shifting (lead-time offset), and creating order proposals and planned orders, is provided by the MM module.
- Lead-time scheduling and capacity planning (capacity evaluation and leveling) use functions from the PP module (production planning).

Procurement and Logistics Execution - Manufacturing execution

- Manufacturing execution is another comprehensive part of product development and manufacturing, supporting not only workshop and continuous flow production but also many other forms, such as make-to-order production, make-to-stock production, repetitive and batch manufacturing, and even process production.

Procurement and Logistics Execution - Manufacturing execution (Cont.)

- Special forms of manufacturing organization, such as Kanban, lean production, pull production, and integrated product and process development (iPPE), are also available.
- PP functions are used for manufacturing execution. Integration of other systems, including both SAP and third-party systems, is made possible by complying with standardized interfaces.

Procurement and Logistics Execution - Product development

- Product development is the section that provides support for, among other things, defining a product, determining the requirements of the product, developing the product, and identifying potential suppliers. These tasks require effective cooperation from all parties involved (project managers, developers, designers, engineers, etc.).
- For this reason, collaborative work is explicitly supported through a process called “development cooperation.” Special attention is paid to cross-enterprise product development, including internal and external project teams.

Procurement and Logistics Execution – Life-cycle data management

- Life-cycle data management refers to all aspects of managing and representing product information, such as product structures, variants, drawings, recipes, routings, etc. It also includes additional specifications, like a classification of hazardous substances and goods.

Procurement and Logistics Execution – Life-cycle data management (Cont.)

- Life-cycle data management involves the administration of a large number of different documents. Therefore, a comprehensive document management section is included.
- Because product data can frequently change, life-cycle data management contains dedicated functionality for change and configuration management.
- Data changes are often due to engineering changes, possibly affecting production orders, planned orders, and purchase orders. An order change management function helps identify the affected orders.

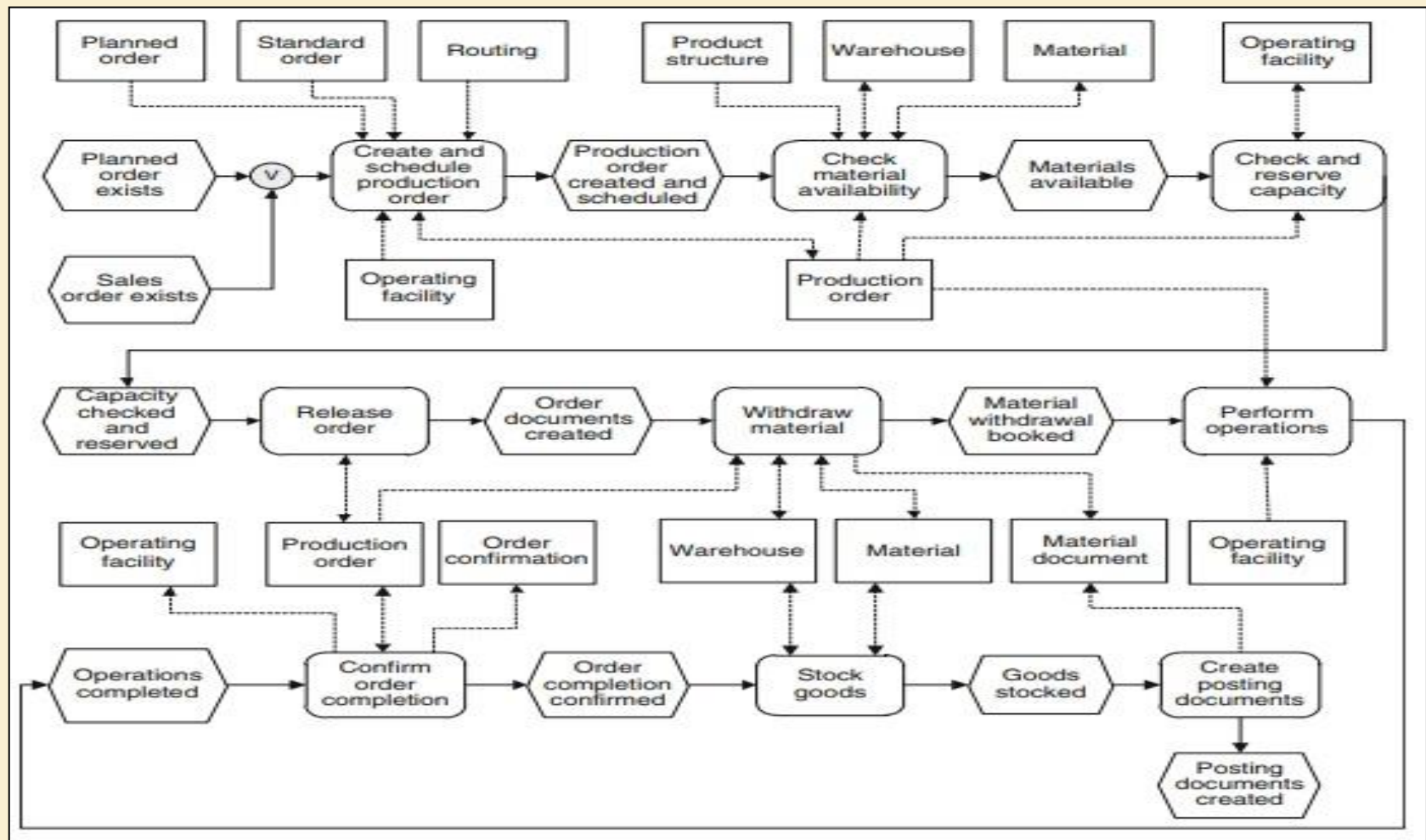
Procurement and Logistics Execution – Production Process

- The production process consists of the various steps and activities involved with the manufacture or assembly of finished goods and semifinished goods.
- A basic production process



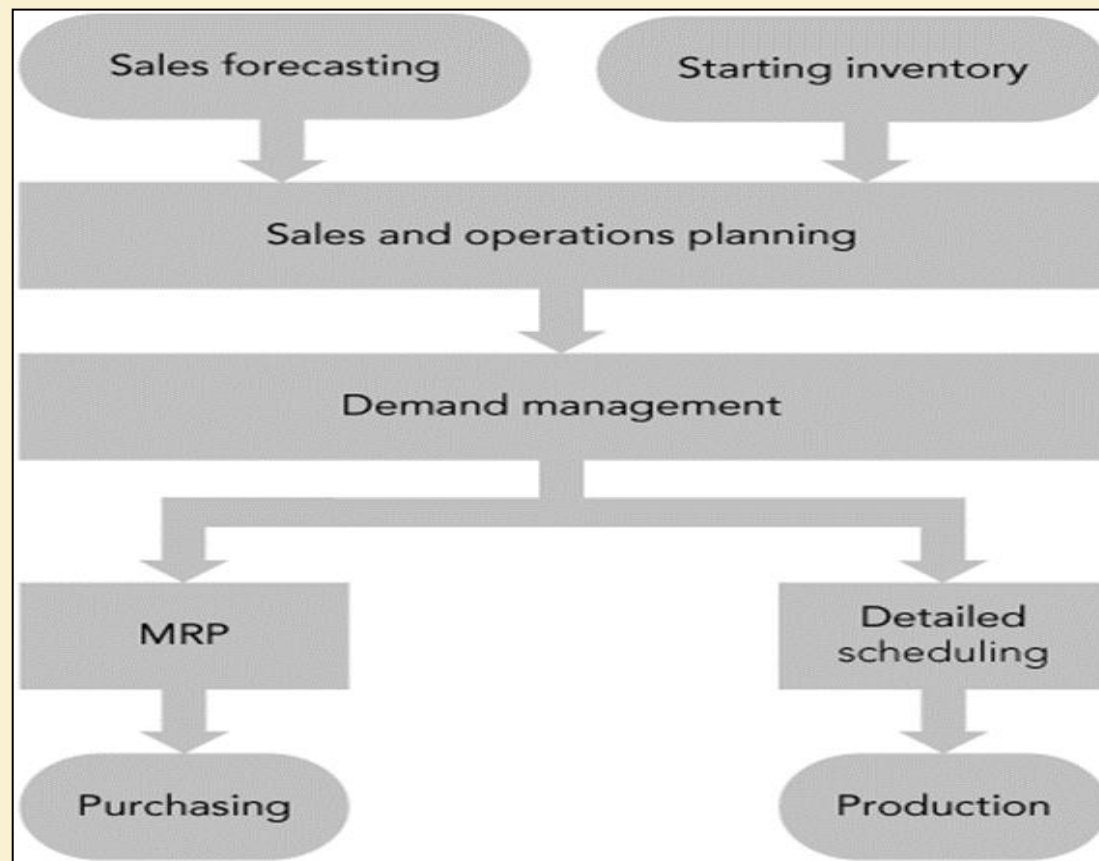
Procurement and Logistics Execution – Production Process (Cont.)

- Production process including information objects.



The SAP ERP Approach to Production Planning

- SAP ERP Approach to Production Planning



In conclusion

- Production Development and Manufacturing.
- Procurement and Logistics Execution.
- The SAP ERP Approach to Production Planning.

Understand



2. MRP - Materials Requirement Planning

What is MRP – Materials Requirement Planning?

- MRP is a system designed to plan manufacturing production.
- It identifies necessary materials, estimates quantities, determines when materials will be required to meet the production schedule, and manages delivery timing with the goal of meeting demands and improving overall productivity.

Master Data for MRP

- The most important master data for production planning and control are data concerning:
 - ❖ Parts.
 - ❖ Product structures.
 - ❖ Operations.
 - ❖ Routings.
 - ❖ Operating facilities or work centers.
 - ❖ Manufacturing structures.

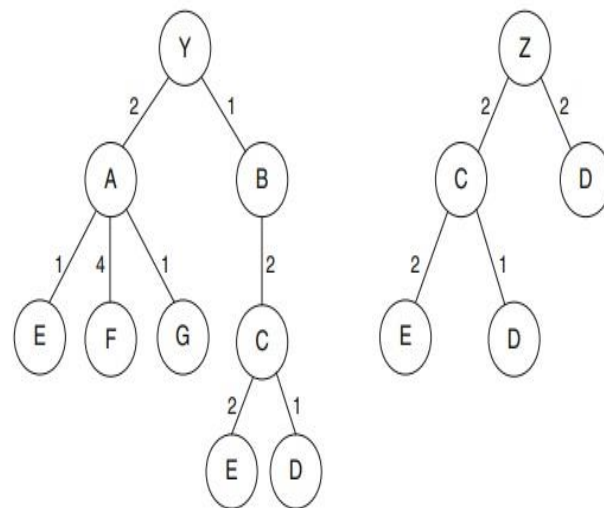
Master Data for MRP – Part and Product Structures

- Part master data plays a central role in every manufacturing application system. The generic term “part” comprises assemblies, component parts, raw materials, end products, and more.
- Considering the number of parts and the number of attributes, part master data is usually quite substantial. Important attributes (or fields) of Part Master Data include the following:
 - ❖ Part number; Variant code; Part name; Part description; Part type (e.g., finished product, assembly, and additional material); Measuring unit (e.g., piece, kg, and m); Form identification; Drawing number; Basic material; Planning type (e.g., in-house production and consumption-driven MRP); Replenishment time; etc.

Master Data for MRP – Part and Product Structures (Cont.)

- Product structures show what parts make up a product. This composition is often depicted as a tree. The edges of the tree represent either “*consists of*” or “*goes into*” relationships, depending on the perspective.
- Product structure trees (“consists of”)

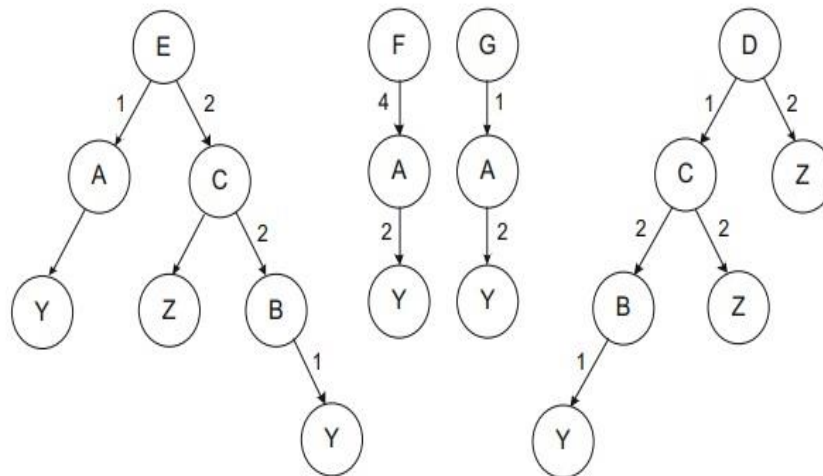
Fig. 2.1 Product structure trees (“consists of”)



The diagram shows two simplified product structure trees for the end products Y and Z. The numbers on the edges are quantity coefficients. Y consists of two units of A and one unit of B. Conversely, A and B go into Y with 2 and 1 units, respectively.

Master Data for MRP – Part and Product Structures (Cont.)

- Product structures show what parts make up a product. This composition is often depicted as a tree. The edges of the tree represent either “*consists of*” or “*goes into*” relationships, depending on the perspective.
- Reversed product structure trees (“goes into”)

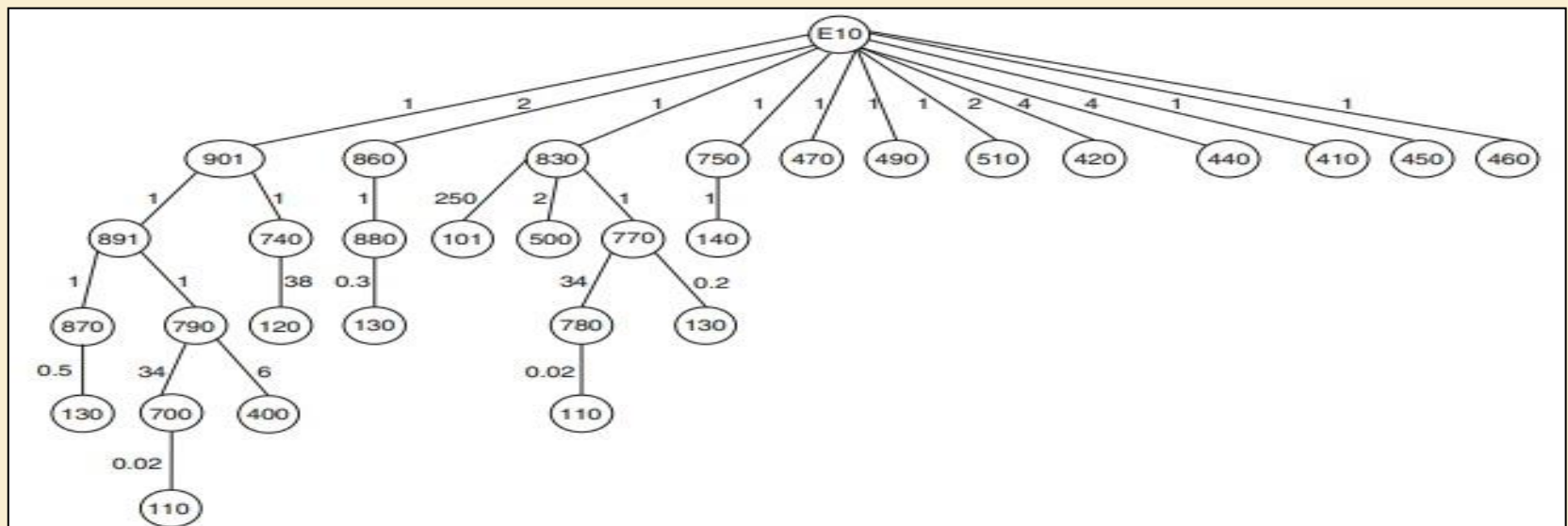


The figure directly shows where a given part is needed. For example, part E goes directly into part A with one unit and into part C with two units, as well as indirectly into parts Z, B, and twice (through parts A and B) into part Y.

Fig. 2.2 Reversed product structure trees (“goes into”)

Master Data for MRP – Part and Product Structures (Cont.)

- Furthermore, there are other perspectives, such as product structure as a Gozinto graph; product structure transformed into a binary tree; and Product structure as a relationship type in an Entity relationship diagram.
- Example of Product structure of an electric motor.



Master Data for MRP – Bill Of Material

- A bill of materials (BOM) represents a product structure together with essential information about the nodes (i.e., part master data) in the form of a list. Each row shows one subordinate part.
- The parts are described by part number, part name, quantity needed for the upper part, etc. In this way, a bill of materials describes the composition of an end product or an intermediate product (assembly).

Master Data for MRP – Bill Of Material (Cont.)

- Bills of materials are employed for various purposes: requirements planning, assembly, computer-aided design, etc.
- The content, structure, and format of a bill of materials depend on the used purpose. Hence, several labels exist, for example, planning BOM, assembly BOM, manufacturing BOM, etc.
- Different types of bills of materials exhibit different structures, depending on how much structural information is mapped to the bill. Relating to this, three types can be determined:

Master Data for MRP – Bill Of Material (Cont.)

- Relating to this, three types can be determined:
 - ❖ **Single-level bills of materials** are used to define the immediate components of a higher-level part, that is, what lower-level parts go directly into the higher-level part.

Fig. 2.8 Single-level BOM for electric motor E10	Single-level Bill of Materials					Page 1
	Part: Electric motor, part-id: E10					
	Part-id	Part name	Unit	Quantity	...	
	901	Case (complete)	pc	1		
	860	Bearing cap with breakout	pc	2		
	830	Arbor (complete)	pc	1		
	750	Base plate 30x40 cm	pc	1		
	510	Junction plate box cap	pc	1		
	490	Junction plate 3-pin	pc	1		
	470	Nut M 4	pc	1		
	460	Rigid coupling Ø 14 mm	pc	1		
	450	Capacitor16 µF	pc	1		
	440	Hex nut M 4x200	pc	4		
	420	Hex nut M 4x10	pc	2		
	410	Hex nut M 8x30	pc	4		

Master Data for MRP – Bill Of Material (Cont.)

- Relating to this, three types can be determined:

❖ **Multilevel bills of materials**, unlike single-level bills, expand the higher-level part down to all levels of the product structure. This type of bill displays the entire product structure tree as a list. The upper-part/lower-part relationships are indicated with level numbers.

Fig. 2.9 Multilevel BOM for electric motor E10

Multi-level Bill of Materials					Page 1
Part: Electric motor, Part-id: E10					
Level	Part-id	Part name	Unit	Quantity	...
1	901	Case (complete)	pc	1	
2	891	Case with laminations	pc	1	
3	870	Housing block (aluminum)	pc	1	
4	130	Aluminum bar	kg	0.5	
3	790	Plate packet (complete)	pc	1	
4	700	Stator plate muller	pc	34	
5	110	Electrical sheet coil 200 mm	m	0.02	
4	400	Rivet 4x150 mm	pc	6	
2	740	Stator winding	pc	1	
3	120	Copper wire Ø 0.5 mm	m	38	
1	830	Arbor (complete)	pc	1	
2	770	Muller plate packet (complete)	pc	1	
3	780	Muller plate	pc	34	
4	110	Electrical sheet coil 200 mm	m	0.02	
3	130	Aluminum bar	kg	0.2	
2	500	Roller bearing	pc	2	
2	101	Round bar 37x30 mm	pc	250	
1	860	Bearing cap with breakout	pc	2	
2	880	Bearing cap (aluminum)	pc	1	
3	130	Aluminum bar	kg	0.3	
1	750	Base plate 30x40 cm	pc	1	
2	140	Sheet metal board St 37	pc	1	
1	510	Junction plate box cap	pc	1	
1	490	Junction plate 3-pin	pc	1	
1	470	Nut M 4	pc	1	
1	460	Rigid coupling Ø 14 mm	pc	1	
1	450	Capacitor 16 µF	pc	1	
1	440	Hex nut M 4x200	pc	4	
1	420	Hex nut M 4x10	pc	2	
1	410	Hex nut M 8x30	pc	4	

Master Data for MRP – Bill Of Material (Cont.)

- Relating to this, three types can be determined:

❖ **Summarized bills of materials** indicate all parts that go into a product but do not reflect the structure of the product. This means that the tree is “compressed” into one level. When a part appears more than once in the product structure, its quantities are added. Consequently, the bill shows only the total quantity needed for one unit of the top part (e.g., the end product).

Fig. 2.10 Summarized BOM for electric motor E10

Summarized Bill of Materials					Page 1
Part: Electric motor, Part-id: E10					
Part-id	Part name	Unit	Quantity	...	
901	Case (complete)	pc	1		
891	Case with laminations	pc	1		
880	Bearing cap (aluminum)	pc	2		
870	Housing block (aluminum)	pc	1		
860	Bearing cap with breakout	pc	2		
830	Arbor (complete)	pc	1		
790	Plate packet (complete)	pc	1		
780	Muller plate	pc	34		
770	Muller plate packet (complete)	pc	1		
750	Base plate 30x40 cm	pc	1		
740	Stator winding	pc	1		
700	Stator plate muller	pc	34		
510	Junction plate box cap	pc	1		
500	Roller bearing	pc	2		
490	Junction plate 3-pin	pc	1		
470	Nut M 4	pc	1		
460	Rigid coupling Ø 14 mm	pc	1		
450	Capacitor 16 µF	pc	1		
440	Hex nut M 4x200	pc	4		
420	Hex nut M 4x10	pc	2		
410	Hex nut M 8x30	pc	4		
400	Rivet 4x150 mm	pc	6		
140	Sheet metal board St 37	pc	1		
130	Aluminum bar	kg	1.3		
120	Copper wire Ø 0.5 mm	m	38		
110	Electrical sheet coil 200 mm	m	1.36		
101	Round bar 37x30 mm	pc	250		

Disadvantage of MRP

- The outcome of material requirements planning (MRP) is quantity—primary and secondary requirements assigned to different periods.
- The main disadvantage of the MRP approach is that it is not certain whether the requirements can be fulfilled because the manufacturing capacities are not taken into consideration.
- To create a possible plan, material requirements planning has to be augmented with *capacity planning and scheduling*. As a result, it is necessary to consider the planning approach of manufacturing resource planning (MRP II).

In conclusion

- What is MRP – Materials Requirement Planning?
- Master Data for MRP.
- Disadvantage of MRP.

Understand



3. MRP II - Manufacturing Resource Planning

What is MRP II – Manufacturing Resource Planning?

- MRP II is a computer-based system that can create detailed production schedules using real-time data to coordinate the arrival of component materials with machine and labor availability.
- MRP II is used widely by itself, but it's also used as a module of more extensive enterprise resource planning (ERP) systems.

Master Data for MRP II

- Master data for manufacturing resource planning, in addition to those used in material requirements planning, includes routings, operating facilities, factory calendars, shift models, tools, and employees.

Master Data for MRP II - Routings

- A routing is a list of operations required for the manufacturing of an in-house-produced part, such as processing times, setup times, operating facilities, and other resources that might be necessary to perform the operations.
- A routing printed on paper or displayed on a screen usually has a header and a body.
 - ❖ The header contains data such as:
 - Routing number; Part the routing refers to; Parts processed in the operations of the routing; Organizational data (e.g., date of creation, date of last modification, and person in charge), etc.
 - ❖ The body is the main part of the route. It consists of the operations required to manufacture the part.

Master Data for MRP II – Routings (Cont.)

- A simplified example of a routing

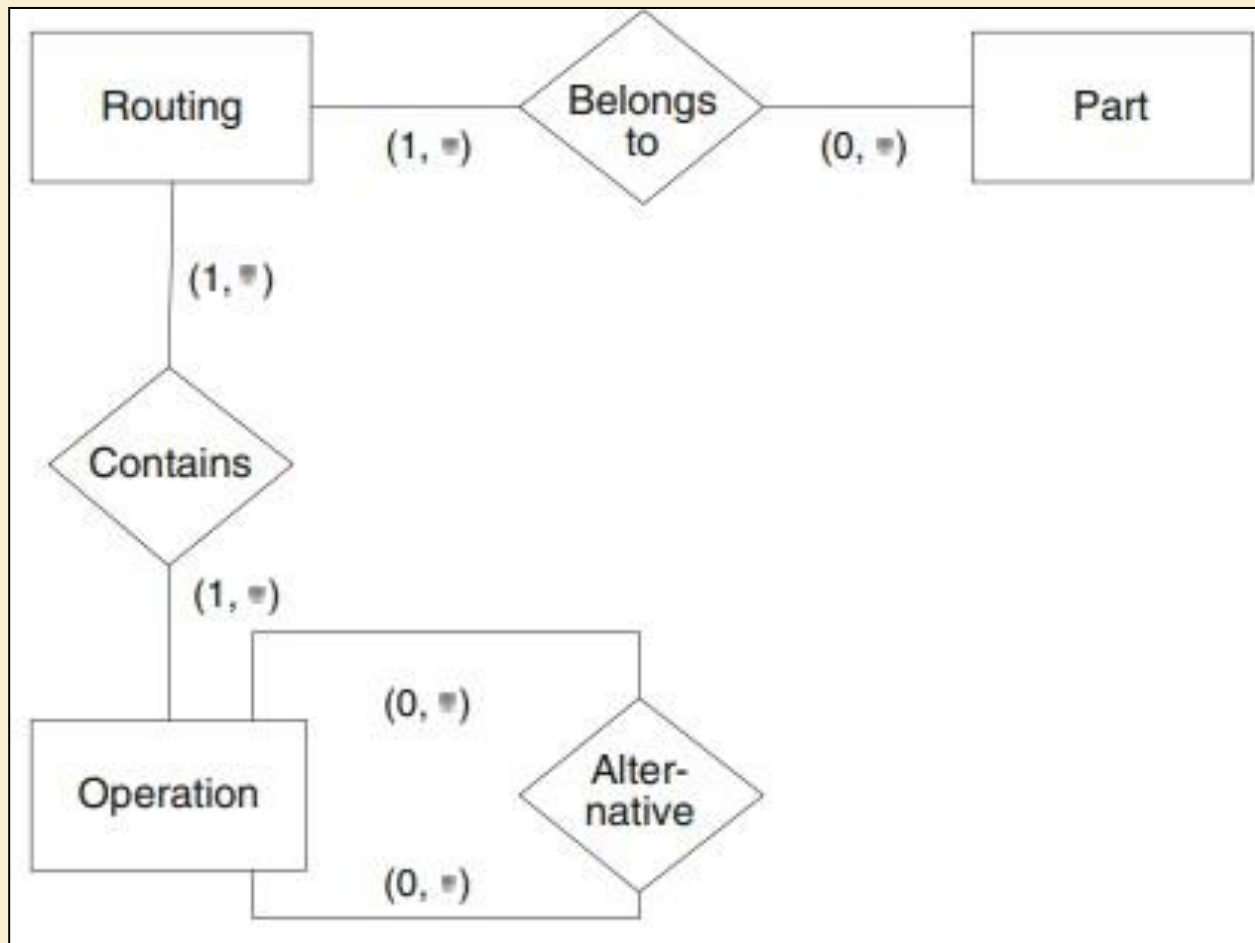
Fig. 3.1 Example of a routing

Routing						Page 1
Part: Bearing cap with breakout, part no: 860						
Material: Bearing cap (aluminum), part no: 880						
Created: 12/10/2012, E. Meier						
Operation no	Operation description	Operating facility	Setup time	Processing time	Drawing no	
6200	Setup lathe	D-40	5			
6300	Mount bearing cap	D-40		3		
6400	Lathe bearing cap according to drawing	D-40		16	31	
6500	Lathe axle breakout according to drawing	D-40		4	32	
6600	Remove bearing cap	D-40				
6700	Mount bearing cap	B-41		2		
6800	Drill fixing hole	B-41		2	33	
6900	Remove bearing cap	B-41				

A simplified example of a routing can be seen in Fig. 3.1. It shows the routing for part number 860 (“bearing cap with breakout”), contained in the product structure of the electric motor E10.

Master Data for MRP II – Routings (Cont.)

- Entity-relationship diagram for routings



The diagram shows a data model for routings that considers the above-mentioned aspects.

It includes the fact that one part may have several routings and that alternatives to an operation may exist.

Master Data for MRP II - Operating Facilities

- Operating facility data are also needed in accounting. For example, the hourly cost rates of the machines are usually stored with the operating facility master data.
- Attributes used to describe operating facilities usually include the following:
 - ❖ Operating facility number; Name and/or description; Location; Cost center; Technical data (e.g., kW and voltage); Capacity (e.g., hours or units per shift and number of shifts); Worker data (e.g., skills required, and number of operators needed); Usage/performance rates; Average setup time; Machine cost rate (€/h); Maintenance data (e.g., maintenance intervals and average downtime); Person in charge; etc.

Master Data for MRP II - Operating Facilities (Cont.)

- Operating facility data is primarily needed for capacity requirement planning. To connect operations and operating facilities, the relationships between the two have to be maintained in the database.

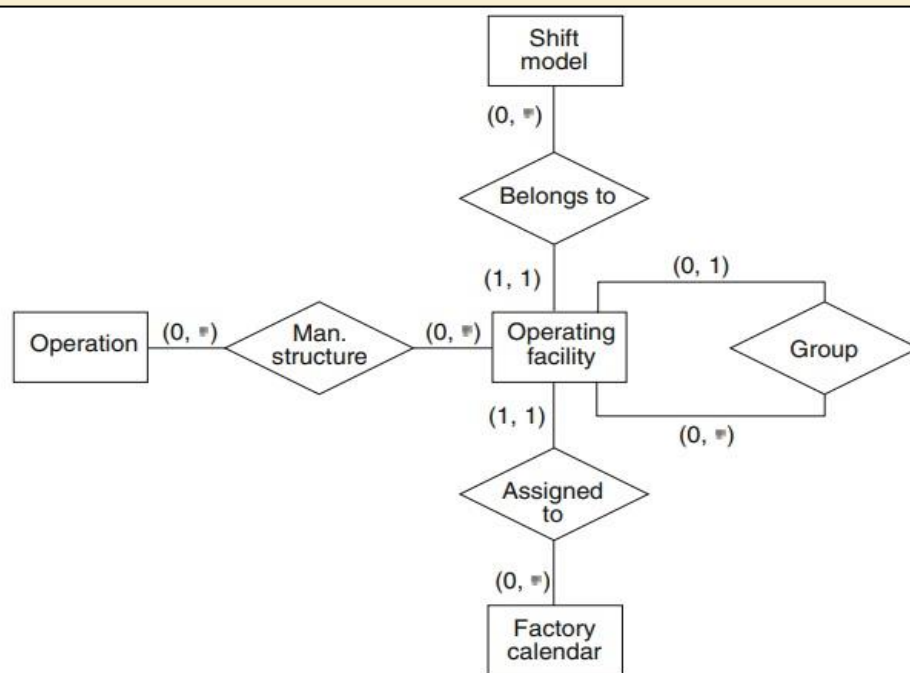


Fig. 3.4 Entity-relationship diagram for manufacturing data

In the diagram, the relationship type “man. structure” (manufacturing structure) is used for this purpose.

The figure also shows how operating facilities can be combined into groups.

Following this approach, an operating facility is either a single facility or a group.

Master Data for MRP II - Factory Calendar

- The factory calendar includes calendar days, hours, and minutes that play an important role in planning, scheduling, and controlling.
- The factory calendar indicates which days are working days and which days are off days (weekends, holidays).
- Many companies have their calendar numbering the working days in a year. The year is not necessarily the calendar year but can also be the company's fiscal year.

Master Data for MRP II – Shift Mode

- A shift model describes the daily work times, specifying the beginning and end of a shift and the breaks during the shift.
- When a company's operation requires the use of several shifts, different shift models may apply to different sectors of the company or, in particular, to different operating facilities.
- The entity-relationship diagram for manufacturing data shows how this requirement is mapped to the data model. The relationship type "*assigned to*" allows different shift models to be assigned to different operating facilities. On the other hand, each facility has a unique shift model.

Master Data for MRP II – Employees

- Employee data belongs primarily to the human resources function. However, they are also needed in manufacturing resource planning.
- The main reason is that in shop-floor control, employees with certain skills may also need to be scheduled.
- Many workplaces require specialists with certain qualifications. The availability of a specialist can impose the same (or even stricter) restrictions as the available capacity of the machine.

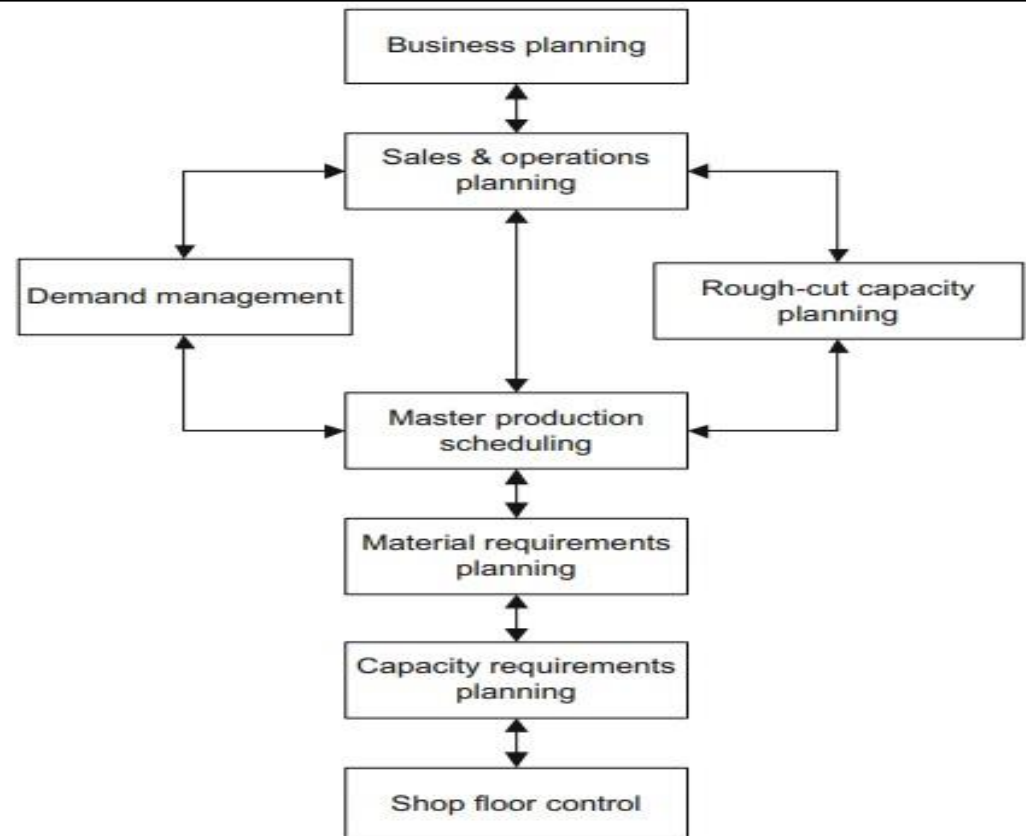
Master Data for MRP II – Tools/Attachment

- Restrictions for scheduling and shop-floor control can also arise from the availability of tools and attachments.
- This is particularly true when special tools are needed in several places. In this case, they must be scheduled, just like other resources (machines, employees, etc.).
- Therefore, tools and attachments are often explicitly stored as particular database objects. In other solutions, they are treated like operating facilities, that is, stored as a type of operating facility.

Planning and Control in MRP II

- The MRP II stages shown in the below diagram are the same as the ones supported by today's ERP systems.

Fig. 3.7 Manufacturing resource planning (MRP II)



Planning and Control in MRP II (Cont.)

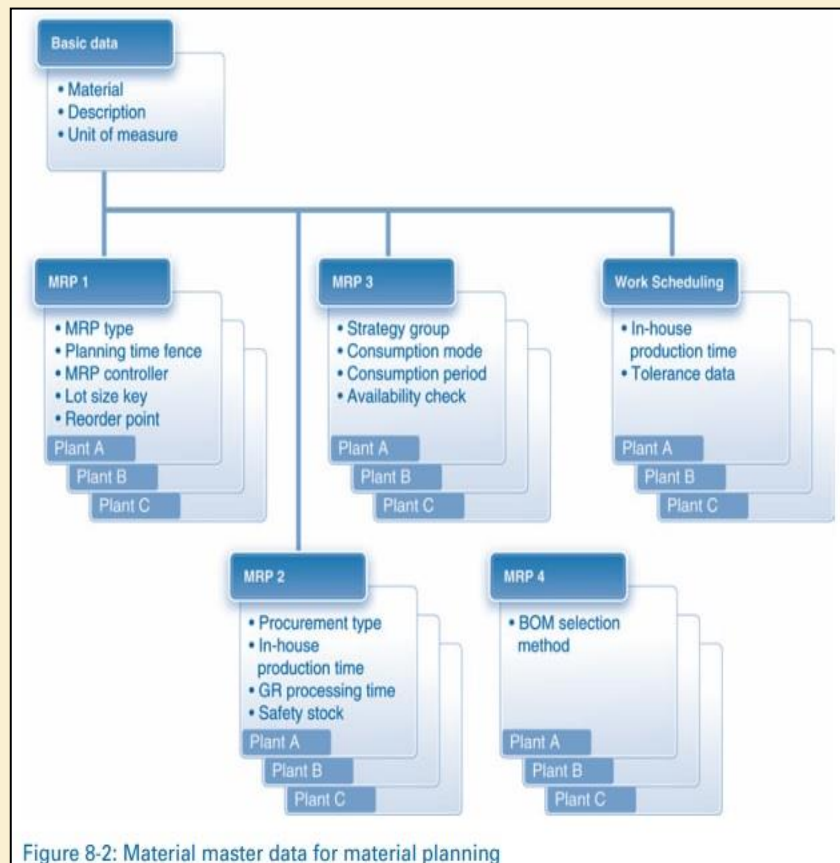
- *The first stage (business planning)*, however, and thus support for the top management, is not at the core of most ERP systems.
- A master production schedule is determined based on long-term *sales and operations planning*.
 - ❖ *Master production scheduling* may have two levels: one being the level of product groups (aggregate production planning) and the other being the level of individual products.

Planning and Control in MRP II (Cont.)

- The capacities needed to produce the primary and secondary requirements are planned in detail in *capacity requirements planning* (CRP).
- In the *shop-floor control* (SFC) stage, the manufacturing orders that are due shortly are released and carried out.
 - ❖ Tasks required for completing the orders include creating order-specific routings, withdrawing the necessary materials from the warehouse, and scheduling the operations at the operating facilities and workplaces.

Other Extensive MRPs

- MRP data can be quite extensive. Consequently, they are divided into four views or tabs, such as MRP 1, MRP 2, MRP 3, and MRP 4, to make the data more readable.



- ❖ The MRP 1 view defines the overall planning strategy used for the material and determines how much material the company should procure.
- ❖ The MRP 2 view identifies the system time that can be used for scheduling and conveys data that the system uses to determine how to procure materials (make vs. buy).
- ❖ The MRP 3 view identifies the strategy that the system will use to calculate how much material is available; as a result, it determines how the material will be produced.
- ❖ The MRP 4 view contains data that the system uses to select the correct BOM.

In conclusion

- What is MRP II – Manufacturing Resource Planning?
- Master Data for MRP II.
- Planning and Control in MRP II.
- Other Extensive MRPs.

Understand



4. Production Planning Master Data

Production Planning Master Data

- Material.
- Bill of Materials (BOM).
- Routing
 - ❖ BOM and routing are like a cooking recipe.
 - ❖ BOM = ingredients.
 - ❖ Routing = steps in the recipe.
- Work Center.
- Product Group.

Production Planning Master Record - Material Master Record

Display Material DXTR1000 (Finished Product)

Other Material

Additional Data

Org. Levels

More

Exit

Sales text

MRP 1

MRP 2

MRP 3

MRP 4

Advanced Planning

Forecasting

W

Material: DXTR1000

Descr.: Deluxe Touring Bike (black)

Plant: DL00 Plant Dallas

General Data

Base Unit of Measure: EA each

MRP Group:

Purchasing Group:

ABC Indicator:

Plant-sp.matl status:

Valid from:

MRP Procedure

Advanced Planning

MRP Type:

Reorder Point:

Planning cycle:

Lot size data

Lot Sizing Procedure:

Minimum Lot Size:

Assembly scrap (%):

Rounding Profile:

MRP areas

Display Material DXTR1000 (MRP 1, Finished Product)

MaterialDXTR1000

Deluxe Touring Bike (black)

Industry sector M Mechanical engineering

Material type FERT Finished Product

Low-level code 000

Created byBOETTCHERon24.05.16

Last changed byBRUSSon27.09.17

Status information:

No deletion flags or locks exist

Production Planning Master Record – Bill of Materials (BOM)

- List of components that make up a product or assembly.
- Wheel Assembly.
 - ❖ Tire
 - ❖ Tube
 - ❖ Wheel
 - ❖ Hex Nut
 - ❖ Lock Washer
 - ❖ Socket Head Bolt
 - ❖ Seat Kit.
 - ❖ Handle Bar.
 - ❖ Pedal Assembly.
 - ❖ Chain.
 - ❖ Brake Kit.
 - ❖ Warranty Document.
 - ❖ Packaging.
- Frame.
- Derailleur Gear Assembly.

Production Planning Master Record – Bill of Materials (BOM) (Cont.)

- Single-Level

Material: Men's Off Road Bike

Plant: Plant Dallas

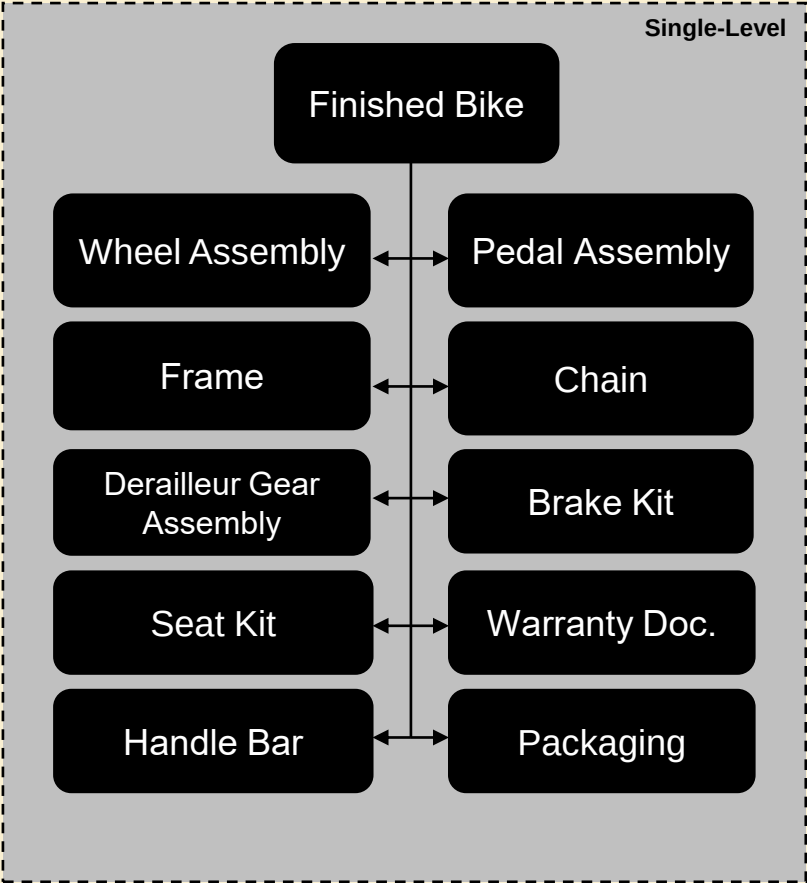
Alternative BOM:

Material

Document

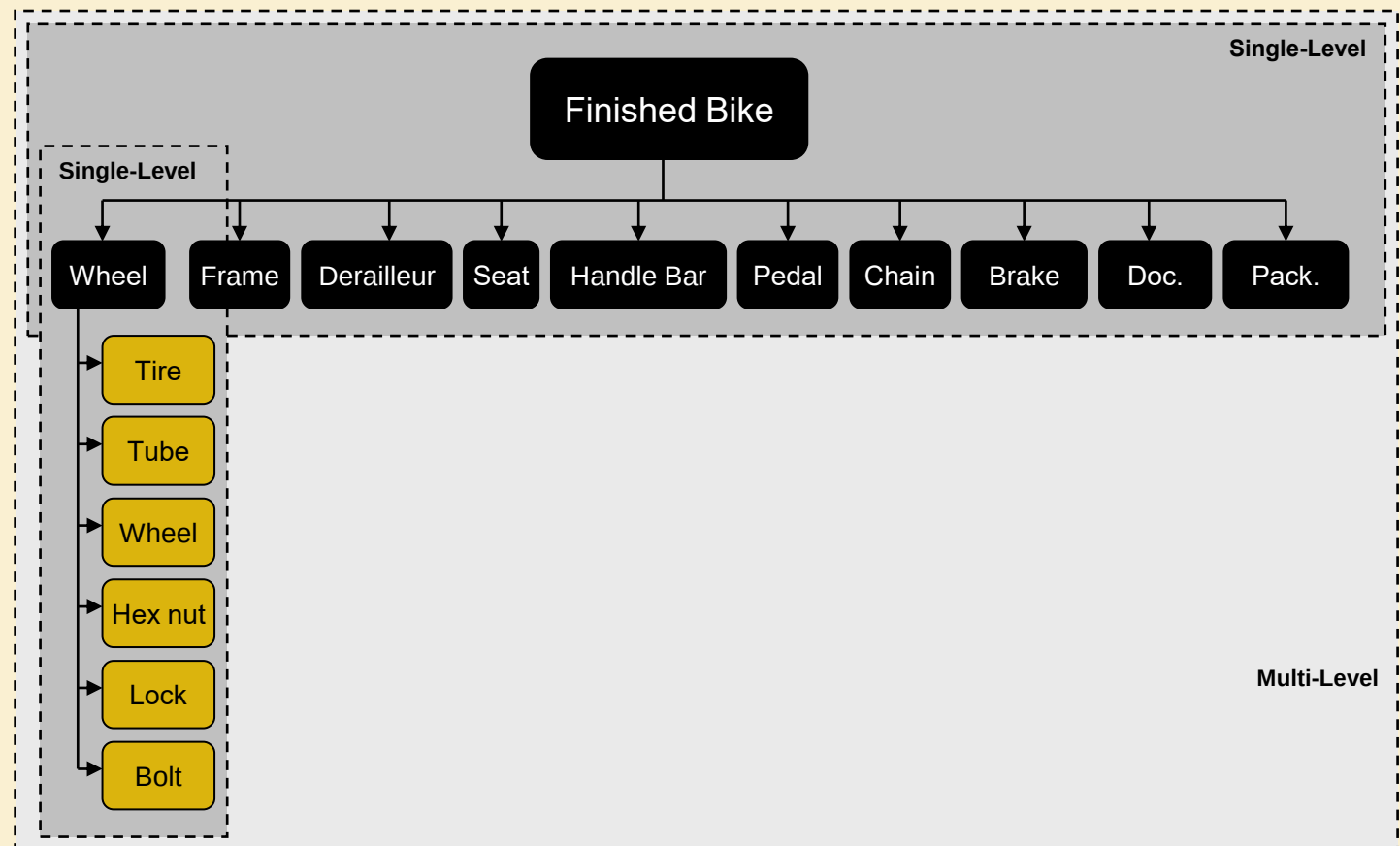
General

☐	Item	ICt	Component	Component description	Quantity	U...
☐	0010	L	ORWA1000	Off Road Aluminum Wheel Assembly	2	EA
☐	0020	L	OFFR1000	Men's Off Road Frame	1	EA
☐	0030	L	DGAM1000	Derailleur Gear Assembly	1	EA
☐	0040	L	ORSK1000	Off Road Seat Kit	1	EA
☐	0050	L	ORHB1000	Off Road Handle Bar	1	EA
☐	0060	L	PEDL1000	Pedal Assembly	1	EA
☐	0070	L	CHAN1000	Chain	1	EA
☐	0080	L	BRKT1000	Brake Kit	1	EA
☐	0090	L	WDOC1000	Warranty Document	1	EA
☐	0100	L	PCKG1000	Packaging	1	EA



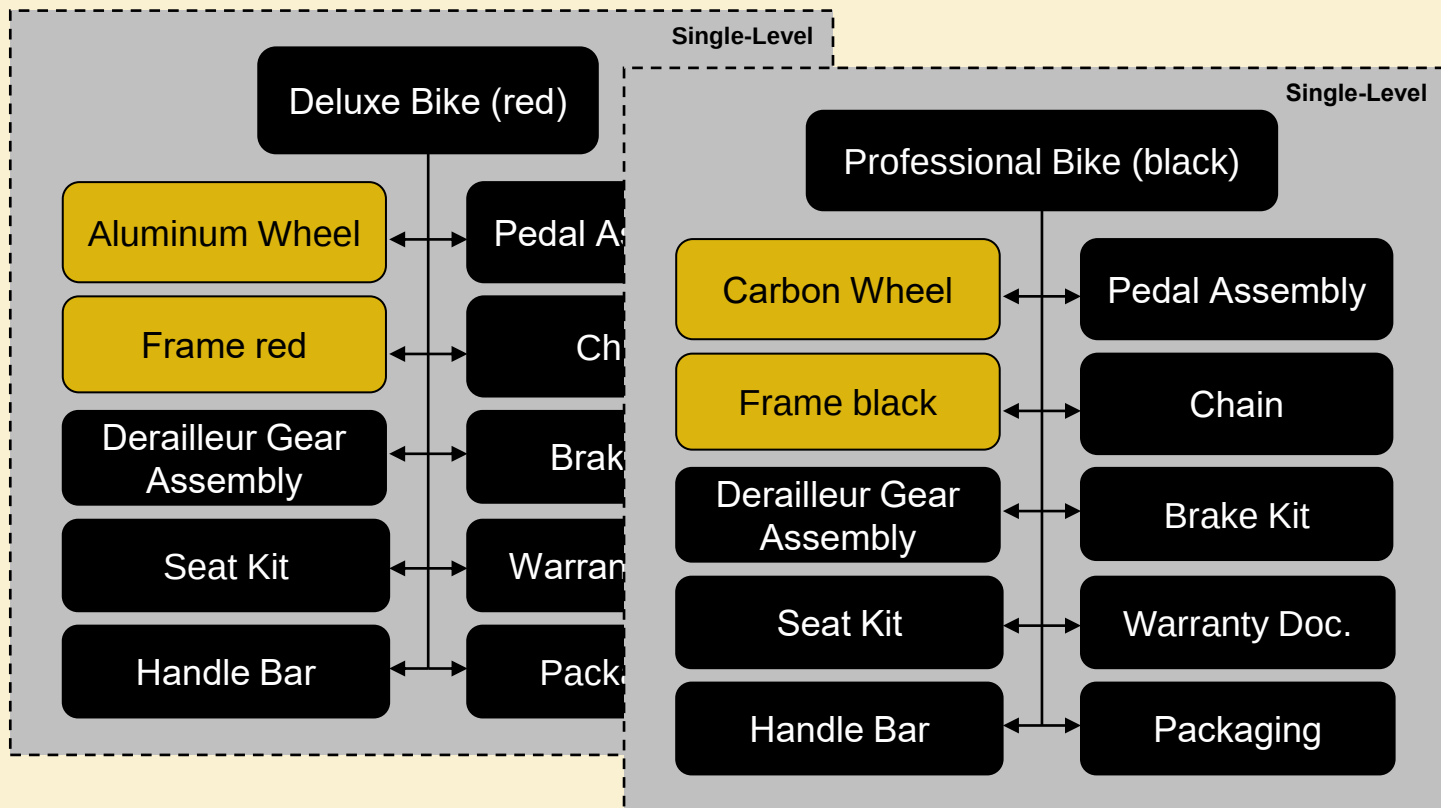
Production Planning Master Record – Bill of Materials (BOM) (Cont.)

- Single-Level vs. Multi-Level



Production Planning Master Record – Bill of Materials (BOM) (Cont.)

- Variant Bill of Materials (BOM)
 - ❖ Several products with a large proportion of identical parts.

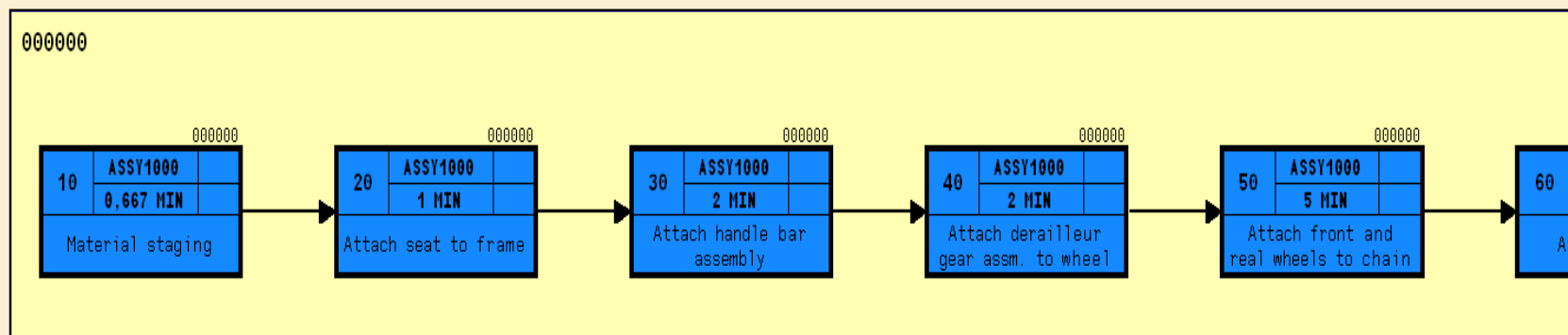


Production Planning Master Record – BOM – Item Categories

- An object that defines items in a BOM according to criteria, such as the object type of the component, for example, material master record or document info record.
- The item category controls the following:
 - ❖ Screen sequence.
 - ❖ Field selection.
 - ❖ Default values.
 - ❖ Material entry.
 - ❖ Inventory management.
 - ❖ Subitems.
- Item Categories:
 - ❖ Stock item.
 - ❖ Non-stock item.
 - ❖ Variable material – Sheet of steel.
 - ❖ Document item.
 - ❖ Text item.

Production Planning Master Record – Routing

- Routings enable the planning of the production of materials (products).
- Routings are used as a template for production orders and run schedules.
- Routings are also used as a basis for product costing.
- Series of sequential steps (operations) that must be carried out to produce a given product.
- Routings contain: What, Where, When, and How.



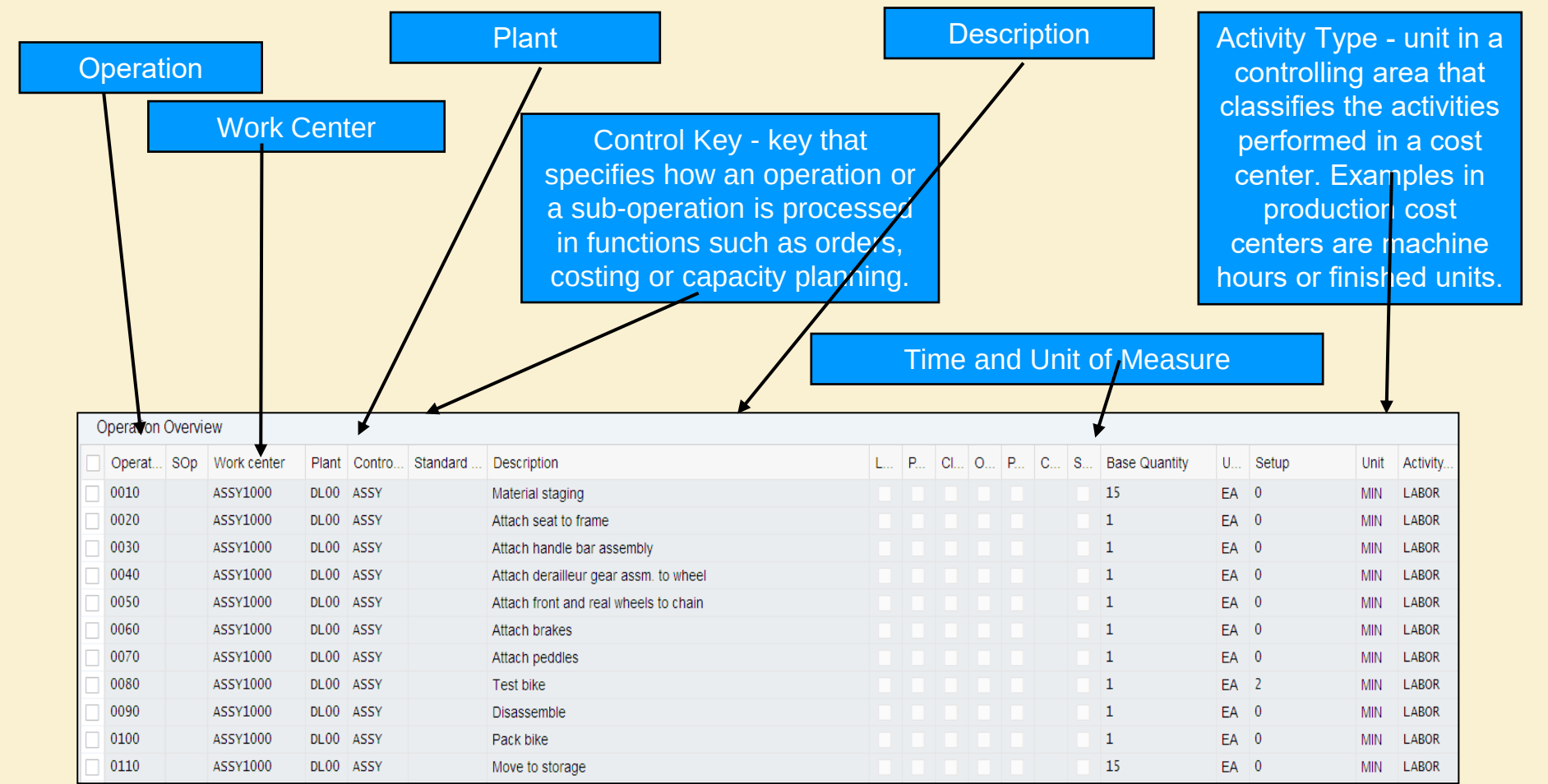
Production Planning Master Record – Routing (Cont.)

- Routing – Operation 20
 - ❖ Attach seat to frame.
- Work Center – ASSY1000
 - ❖ Assembly Work Center.
- Time
 - ❖ 1 minute.

20	ASSY1000	
	1 MIN	
Attach seat to frame		

Production Planning Master Record – Routing (Cont.)

- Routing for Finished Bike



Production Planning Master Record – Work Center

- A location within a plant where value-added work (operations or activities) is performed.
 - ❖ Work Centers can represent
 - People or Groups of people.
 - Machines or Groups of machines.
 - Assembly Lines.
- Work Center used to define capacities
 - ❖ Labor.
 - ❖ Machine.
 - ❖ Output.
 - ❖ Emissions.
- Capacities used in
 - ❖ Capacity requirements planning (CRP).
 - ❖ Detailed scheduling.
 - ❖ Costing.

Production Planning Master Record – Work Center (Cont.)

- Work Centers capture and use the following Resource related data.

❖ ***Basic Data***

- Person Responsible,
Location of Work Center.

❖ ***Scheduling Information***

- Queues and Move Times
(interoperation), Formula
Keys.

❖ ***Costing Data***

- Cost Center, Activity
Types.

❖ ***Personnel Data***

- People, Positions,
Qualifications.

❖ ***Capacity Planning***

- Available Capacity,
Formulas, Operating
Time.

❖ ***Default Data***

- Control Key, Standard
Text Key.

Production Planning Master Record – Product Group

- Aggregate planning that groups together materials or other product groups (Product Families).
- Multi- or Single-Level Product Groups
 - The lowest level must always consist of materials.

In conclusion

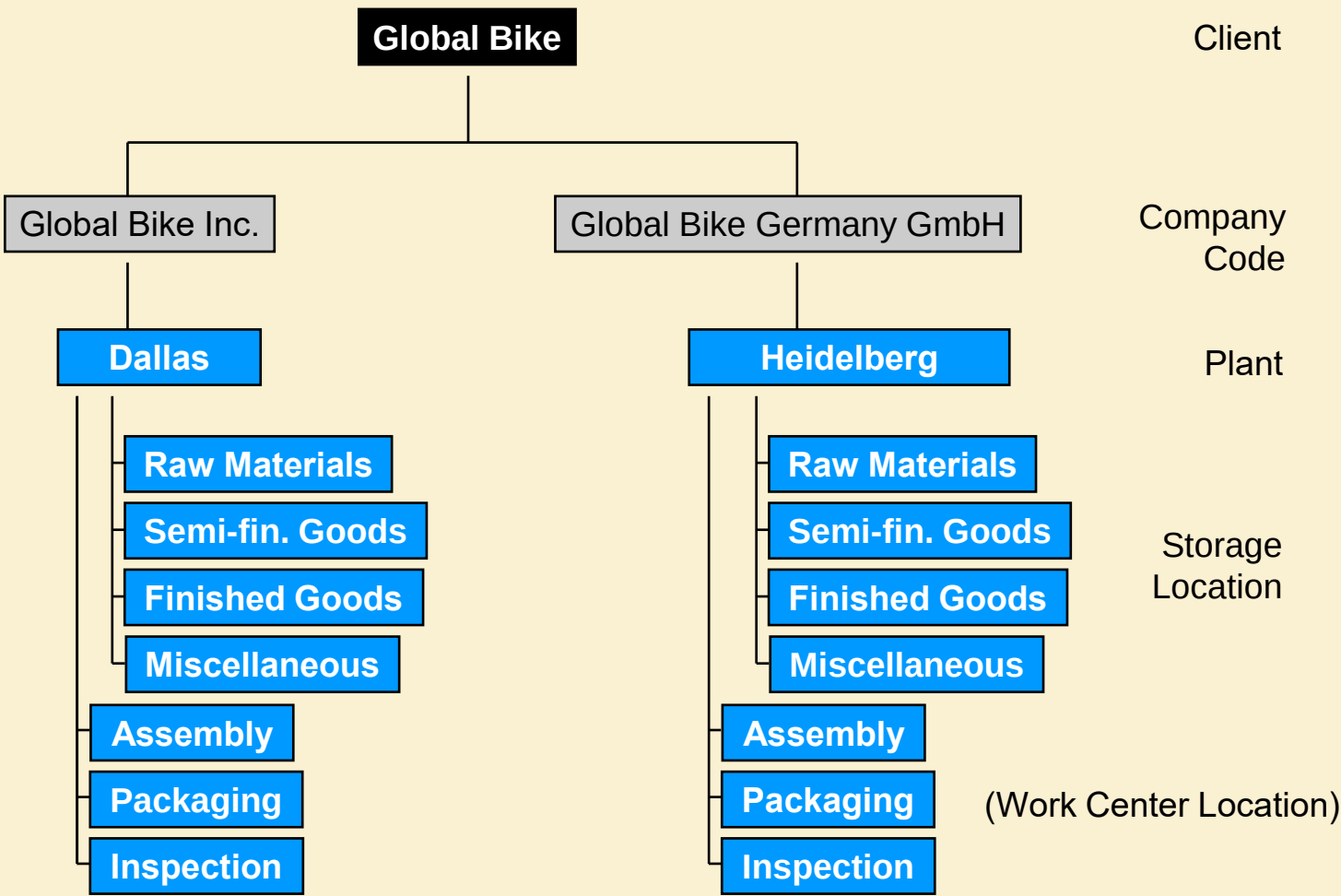
- Material.
- Bill of Materials (BOM).
- Routing
 - ❖ BOM and routing are like a cooking recipe.
 - ❖ BOM = ingredients.
 - ❖ Routing = steps in the recipe.
- Work Center.
- Product Group.

Understand



5. Production Planning Processes

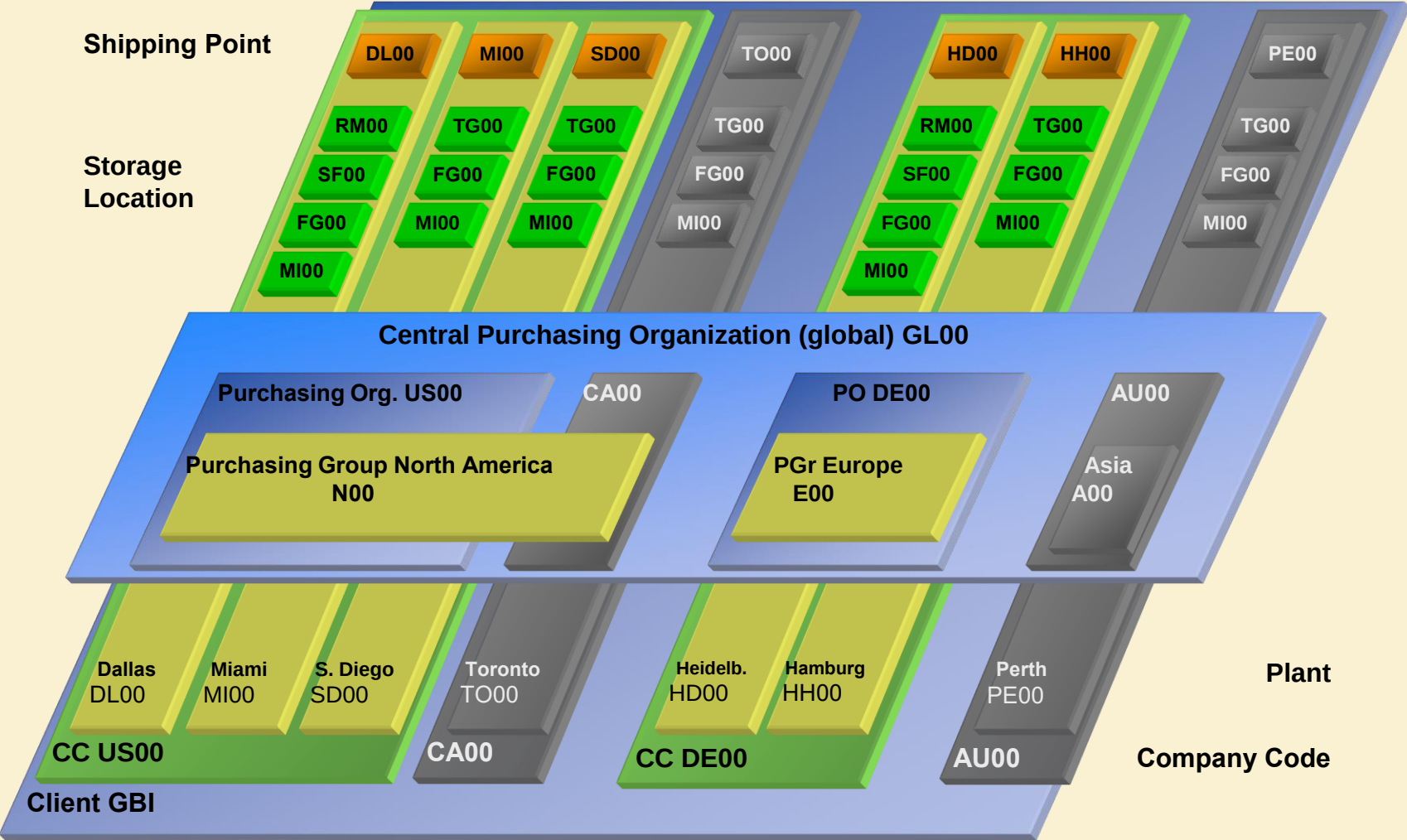
Global Bike Structure for Production Planning



PP Organizational Structure

- Client
 - ❖ An independent environment in the system.
- Company Code
 - ❖ Smallest org unit for which you can maintain a legal set of books.
- Plant
 - ❖ Operating area or branch within a company
 - Manufacturing, distribution, purchasing or maintenance facility.
- Storage Location
 - ❖ An organizational unit allowing differentiation between the various stocks of a material in a plant.
- Work Center Locations (in SAP system → master data)
 - ❖ An organizational unit that defines where and when an operation is performed.
 - ❖ Has an available capacity.
 - ❖ Activities performed are valued by charge rates, which are determined by cost centers and activity types.
 - ❖ Can be machines, people, production lines or groups of tradespeople.

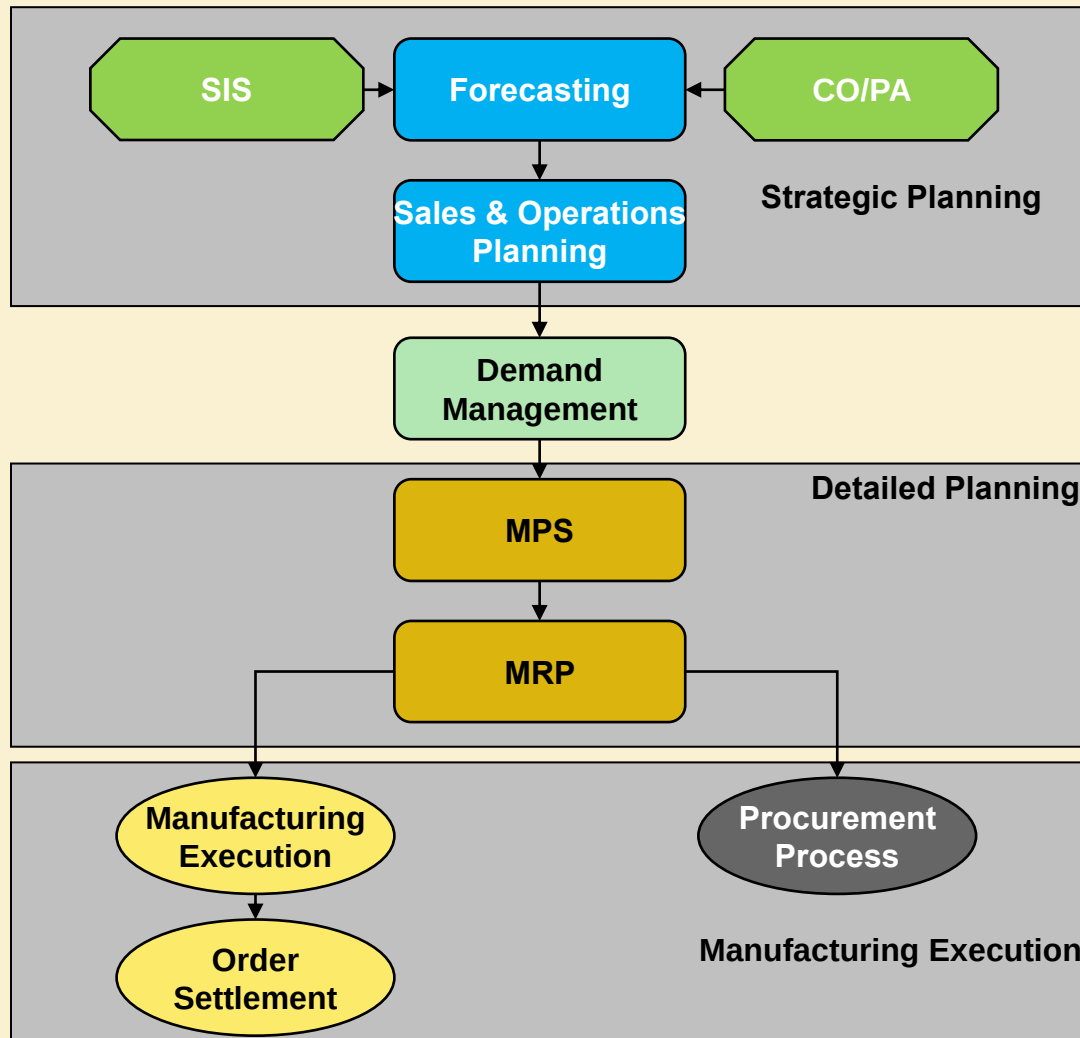
Global Bike Enterprise Structure in SAP ERP (Logistics)



PP Processes

- Production Planning & Execution
 - ❖ Forecasting.
 - ❖ Sales and Operations Planning (SOP).
 - ❖ Demand Management.
 - ❖ Master Production Scheduling (MPS).
 - ❖ Material Requirement Planning (MRP).
- Production Order.

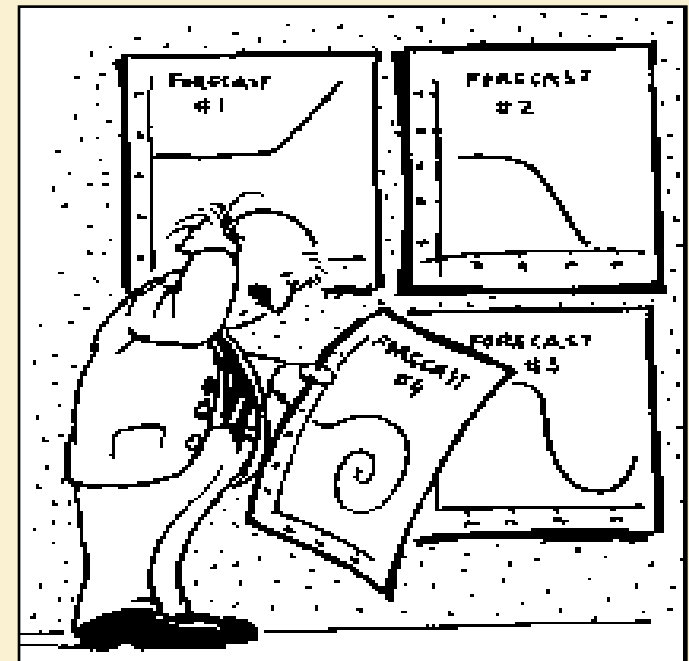
PP Processes – Production Planning & Execution



- **Players in the Game**
 - ❖ **Strategic Planning**
 - CEO, COO, CIO, CFO, Controller, Marketing Director.
 - ❖ **Detailed Planning**
 - Line Managers, Production Scheduler, MRP Controller, Capacity Planners.
 - ❖ **Execution**
 - Line Workers, Shop Floor Supervisors.

Production Planning & Execution - Forecasting

- Forecasting is the foundation of a reliable SOP.
- Accurate forecasts are essential in the manufacturing sector.
- Overstocked & understocked warehouses result in the same issue: A loss in profits.
- Forecasts are ALWAYS WRONG.



Production Planning & Execution – Forecasting (Cont.)

- Forecasting Models
 - ❖ Trend.
 - ❖ Seasonal.
 - ❖ Trend and Seasonal.
 - ❖ Constant.
- Selecting a Model
 - ❖ Automatically.
 - ❖ Manually.

Forecast: Parameters for Automatic Model Selection

Exponential smoothing, first-order with test for

☐ Trend

Alpha factor	0.20
Beta factor	0.20

☐ Season

Alpha factor	0.20
Gamma factor	0.20
Periods per season	12

☒ Trend and season

☐ Seasonal model and test for trend

☐ Trend model and test for season

Alpha factor	0.20
Beta factor	0.20
Gamma factor	0.20
Periods per season	12

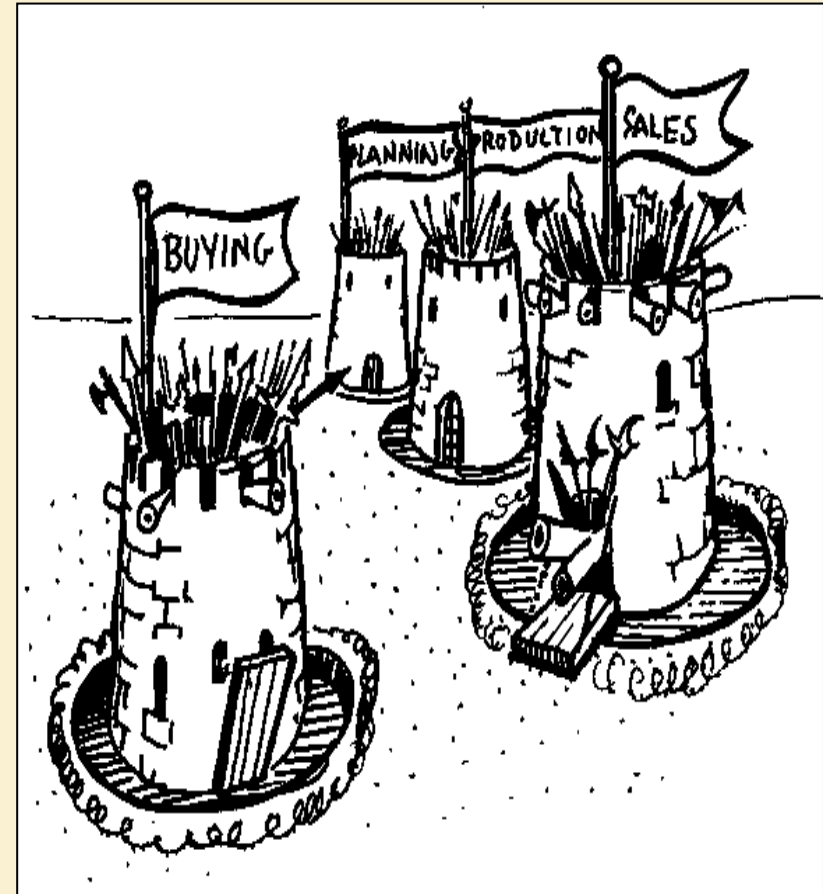
☐ Forecast model sel. using procedure 2

Periods per season	12
--------------------	----

Forecasting Historical... X

Production Planning & Execution – Sales and Operations Planning (SOP)

- Information Origination
 - ❖ Sales.
 - ❖ Marketing.
 - ❖ Manufacturing.
 - ❖ Accounting.
 - ❖ Human Resources.
 - ❖ Purchasing.
- Intra-firm Collaboration
 - ❖ Institutional Common Sense



Sales and Operations Planning (SOP)

- Flexible forecasting and planning tool
- Usually consists of three layers:
 - Sales Plan
 - Production Plan
 - Rough Cut Capacity Plan
- Planned at an aggregate level in time buckets

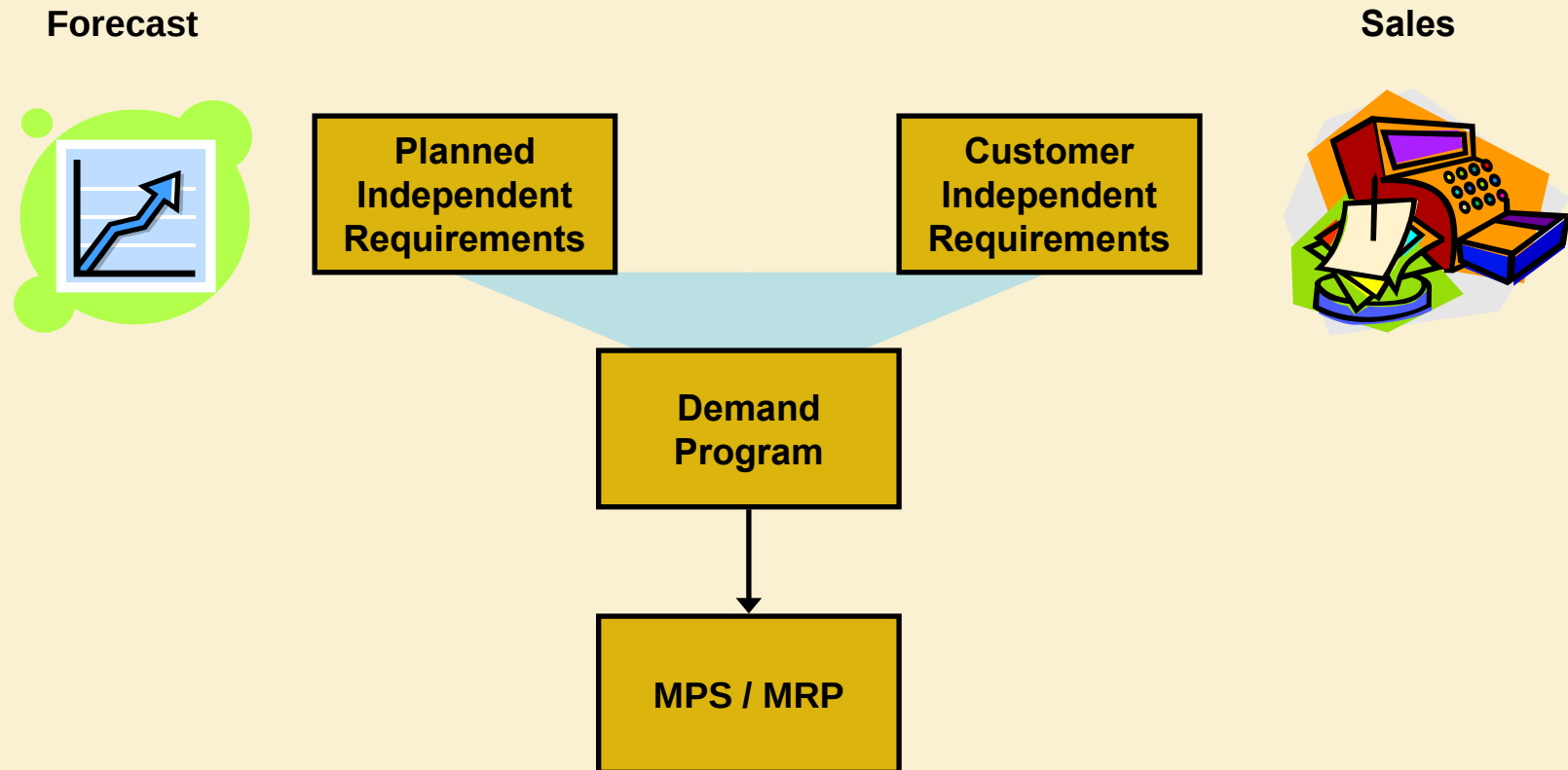
Production Planning & Execution – Demand Management

- The planning of requirement quantities and requirement dates for finished products and important assemblies.
- Definition of the strategy for planning and producing or procuring a finished product.
- Link between Strategic Planning (SOP) & Detailed Planning (MPS/MRP).

Production Planning & Execution – Demand Management (Cont.)

- Demand management can be done manually or based on previous planning results such as sales planning, SOP, and forecast.
- The results of Demand Management is called the Demand Program, it is generated from our independent requirements - PIR - Planned Independent Requirements and CIR - Customer Independent Requirements.

Production Planning & Execution – Demand Management (Cont.)



Production Planning & Execution – Planning Strategies

- Planning strategies represent the business procedures for
 - ❖ The planning of production quantities.
 - ❖ Dates.
- Wide range of strategies.
- Multiple types of planning strategies based upon environment.
 - ❖ Make-To-Stock (MTS).
 - ❖ Make-To-order (MTO).
 - Driven by sales orders.
 - ❖ Configurable materials.
 - Mass customization of one.
 - ❖ Assembly orders.

Production Planning & Execution – Planning Strategy for Make-to-Stock

- Planning takes place using Independent Requirements.
- Sales are covered by make-to-stock inventory.
- Strategies.
 - ❖ 10 – Net Requirements Planning.
 - ❖ 11 – Gross Requirements Planning.
 - ❖ 30 – Production by Lot Size.
 - ❖ 40 – Planning with Final Assembly.

Production Planning & Execution – Planning Strategy for Make-to-Order

- Planning takes place using Customer Orders.
- Sales are covered by make-to-order production.
- Strategies.
 - ❖ 20 – Make to Order Production.
 - ❖ 50 – Planning without Final Assembly.
 - ❖ 60 – Planning with Planning Material.

Production Planning & Execution – Master Production Scheduling (MPS)

- MPS allows a company to distinguish planning methods among materials that have a strong influence on profit or use critical resources and those that do not.

Production Planning & Execution – Material Requirement Planning (MRP)

- In MRP, the system calculates the net requirements while considering available warehouse stock and scheduled receipts from purchasing and production.
- During MRP, all levels of the bill of material are planned.
- The output of MRP is a detailed production and/or purchasing plan.
- Detailed planning level
 - ❖ Execute primary functions.
 - ❖ Monitor inventory stocks.
 - ❖ Determine material needs.
 - Quantity.
 - Timing.
- Generate purchase or production orders.

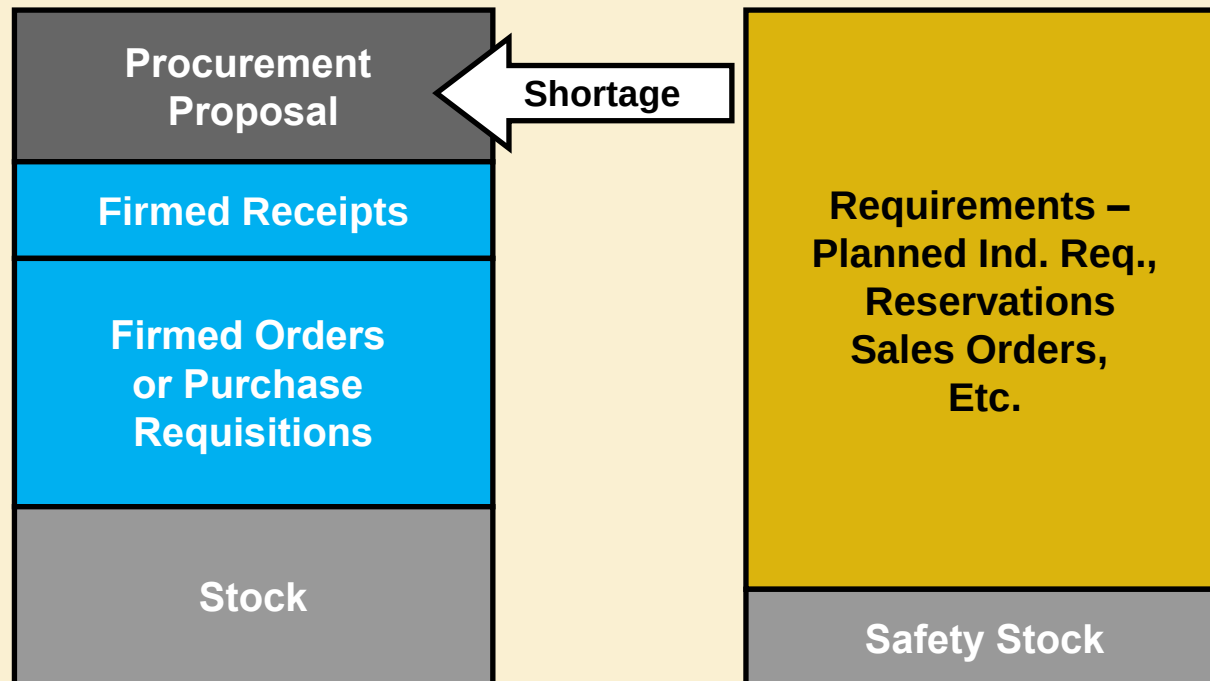
Production Planning & Execution – Demand-Independent vs. Dependent

- Independent Demand – Original source of the demand.
 - ❖ **Independent demand** is demand for a finished product, such as a computer, bicycle, or pizza.
- Dependent Demand – Source of demand resides at another level.
 - **Dependent demand**, on the other hand, is demand for component parts or subassemblies. For example, this would be the microchips in the computer, wheels on the bicycle, or cheese on the pizza.

Production Planning & Execution – Material Requirement Planning (MRP) (Cont.)

- MRP is used to ensure the availability of materials based on the need generated by MPS or the Demand Program.
 - ❖ 5 Logical Steps.
 - Net Requirements Calculation.
 - Lot Size Calculation.
 - Procurement Type.
 - Scheduling.
 - BOM Explosion.

Material Requirement Planning (MRP) – Net Requirements



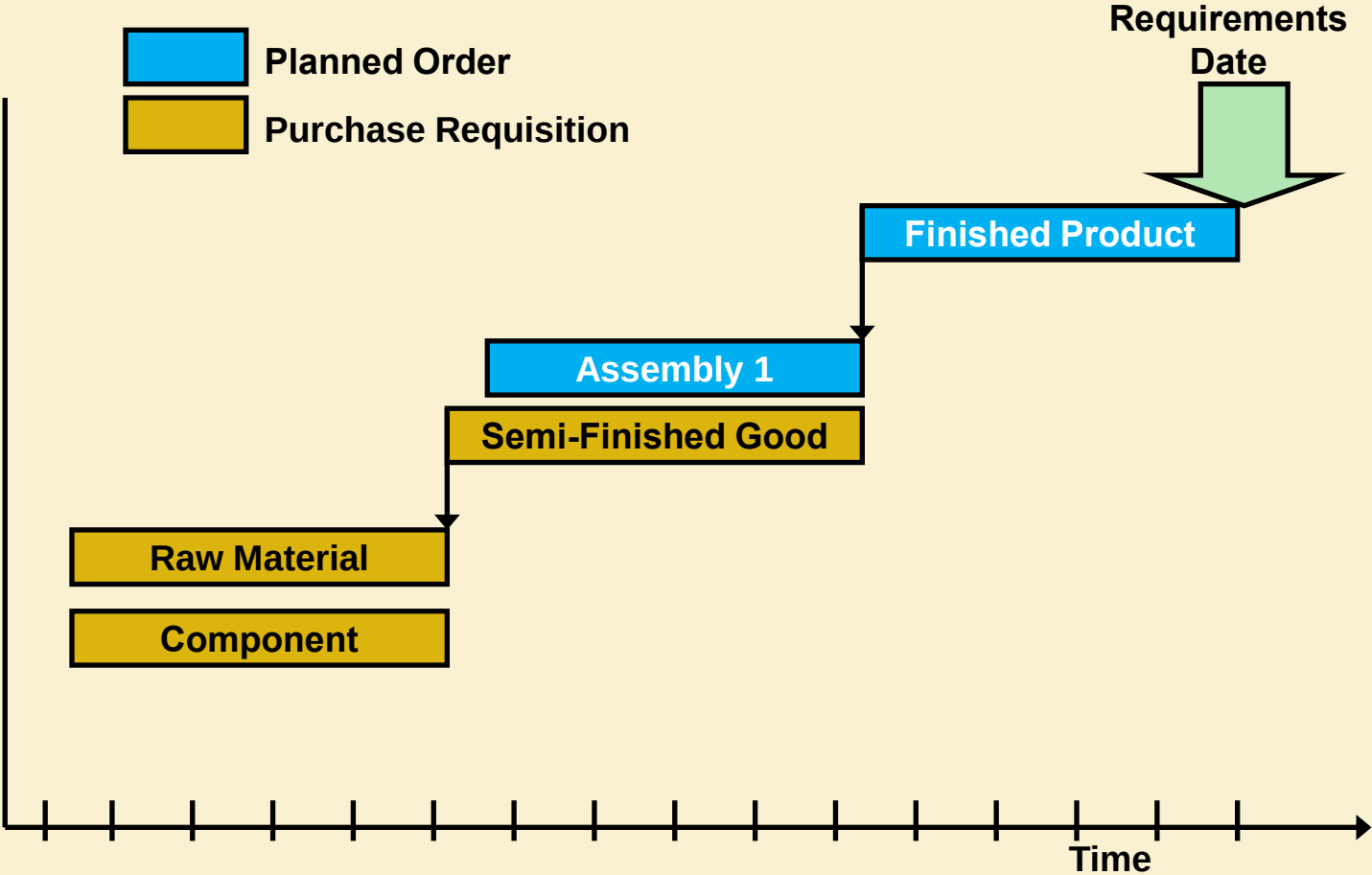
Material Requirement Planning (MRP) – Lot sizing

- Static
 - ❖ Based on fixed values in the Material Master.
- Periodic
 - ❖ Groups net requirements together from multiple periods.
- Optimized
 - ❖ Calculates the optimum lot size for a several periods of net requirements.

Material Requirement Planning (MRP) – Procurement Type

- External Procurement.
 - ❖ Purchase Requisition.
 - ❖ Purchase Order.
 - ❖ Schedule Line.
- Internal Procurement.
 - ❖ Planned Order.
 - ❖ Production Order.
 - ❖ Process Order.

Material Requirement Planning (MRP) – Multi-Level Scheduling

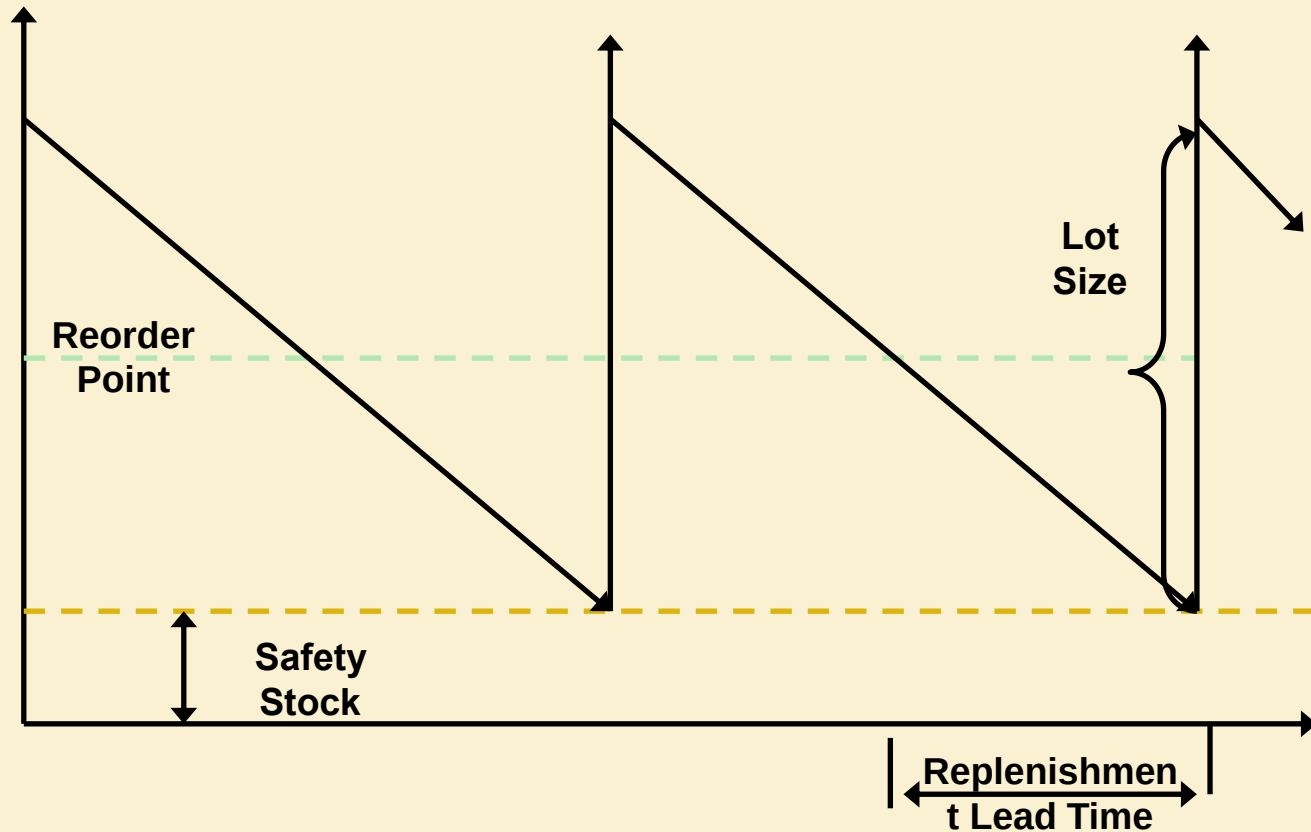


Production Planning & Execution – MRP vs. Consumption-Based

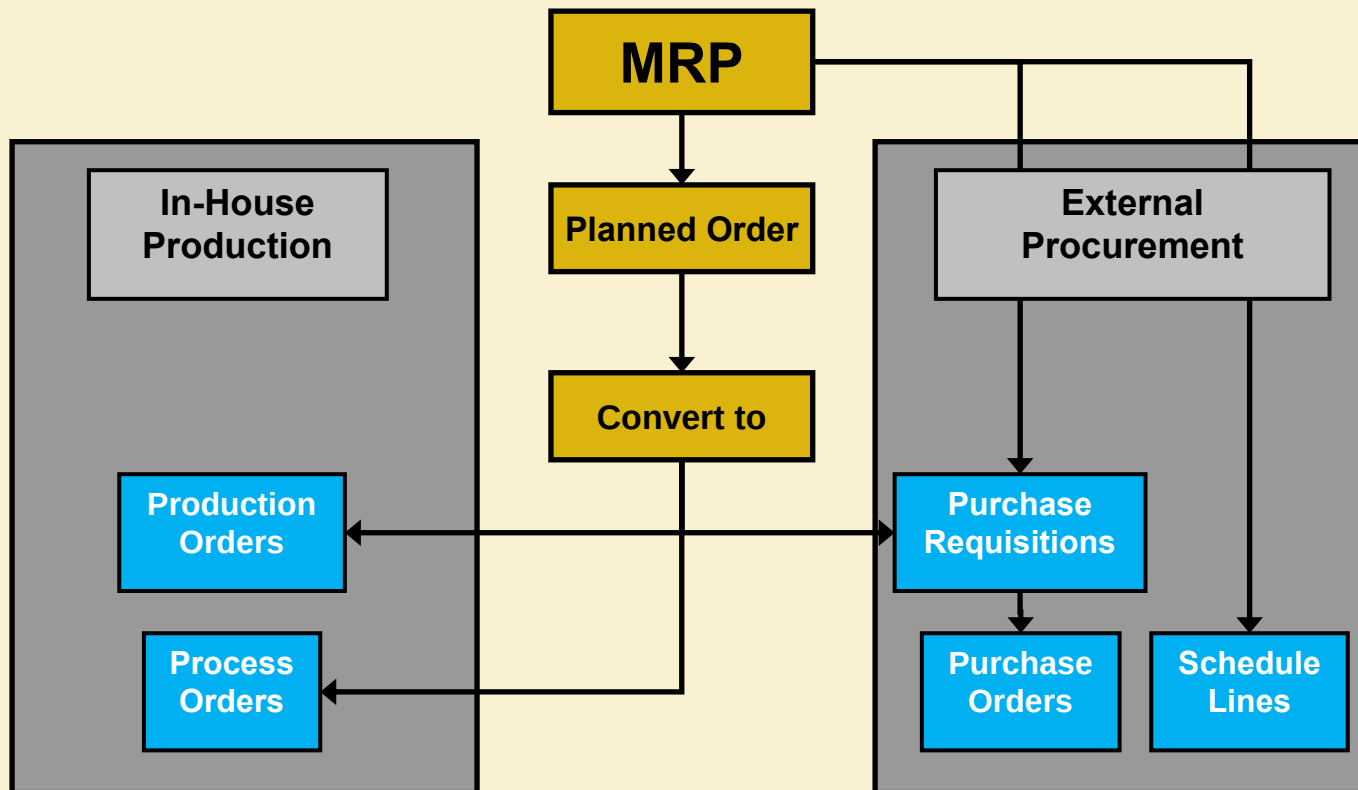
- Whether or not a material is planned using MRP or Consumption Based is determined by the MRP Type on the MRP1 screen of the Material Master.

MRP	Consumption Based
PD – MRP	VB – Reorder-Point
VSD – Seasonal MRP	VV – Forecast Based
	RP – Replenishment

Production Planning & Execution – Consumption-Based



Production Planning & Execution – Output of MRP



PP Processes – Orders, orders, orders

- Planned Order (planning)
 - ❖ A request created in the planning run for a material in the future (converts to either a production or purchase order).
- Production Order (execution)
 - ❖ A request or instruction internally to produce a specific product at a specific time.
- Purchase Order (execution)
 - A request or instruction to a vendor for a material or service at a specific time.

PP Processes – Production Order

- Production orders are used to control production operations and associated costs
 - ❖ Production orders define the following
 - Material produced.
 - Quantity.
 - Location.
 - Timeline.
 - Work involved.
 - Resources used.
 - Cost settlement.

Production Planning & Execution – Production Order (Cont.)

How

What

How many

Components

Timeline

<

SAP

Production order Create: Header

Release Order

Schedule Order

Cost Order

Material

Capacity

Operations

Components

Documents

Sequences

More

Order: %000000000001

Type: PP01

Material: DXTR1000

Deluxe Touring Bike (black)

Plant: DL00

Status: CRTD MSPT RELR SETC

General

Assignment

Goods Receipt

Control

Dates/Qties

Master Data

Long Text

Administration

Items

Fast Entry

Quantities

* Total Qty: 10 EA

Delivered: 0

Scrap Portion: 0,00 %

Short/Exc. Rcpt: 0

Dates/Times

Basic Dates

Scheduled

Confirmed

End: 16.04.2018 00:00

14.04.2018 13:39

Start: 13.04.2018 00:00

14.04.2018 08:00

00:00

Release:

13.04.2018

Scheduling

Floats

* Type: 2 Backwards

Sched. Margin Key: 001

Reduction: No reduction carried out

Float Bef. Prdn: 1 Workdays

Note: Automatically carried out today scheduling

Float After Prdn: 1 Workdays

Priority:

Release Period: 1 Workdays

In conclusion

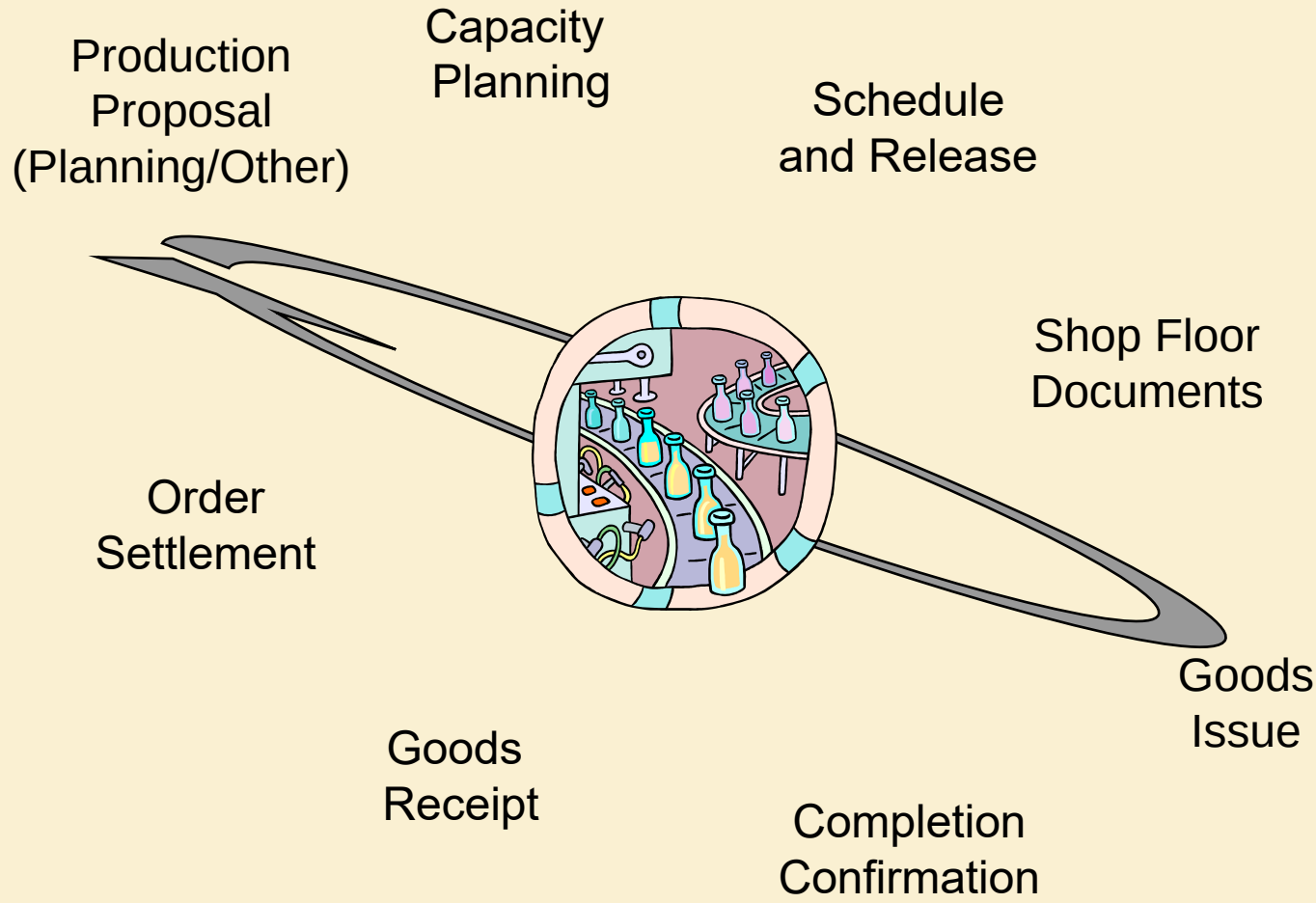
- Production Planning & Execution
 - ❖ Forecasting.
 - ❖ Sales and Operations Planning (SOP).
 - ❖ Demand Management.
 - ❖ Master Production Scheduling (MPS).
 - ❖ Material Requirement Planning (MRP).
- Production Order.

Understand



6. Manufacturing Execution Process

Manufacturing Execution Process



Manufacturing Execution Process - Schedule

- Calculates the production dates and capacity requirements for all operations within an order.
 - ❖ Determines a Routing
 - Operation specific timelines.
 - Material Consumption Points.
 - ❖ Material Master
 - Scheduling Margin Key (Floats).
 - ❖ Work Center
 - Formulas.
 - Standard Inter-operation Times.

Manufacturing Execution Process - Release

- Two release processes.

- ❖ ***Header Level.***

- Entire order and all operations are released for processing, order is given a REL status.

- ❖ ***Operation Level***

- Individual operations within an order are released.
- Order is given a PREL status.
- Not until the last operation is released does the order obtains a REL status.

- Automatic vs. manual.

Manufacturing Execution Process - Availability Check

- Automatic check to determine whether the component, production resource tools, or capacities in an order are available.
 - ❖ Can be automatic or manually executed.
 - ❖ Determines availability on the required date.
- Generates an availability log
 - ❖ Displays results of the check.
 - ❖ Missing parts list.
 - ❖ Reservations that could not be verified.

Manufacturing Execution Process - Schedule & Release

- The time between scheduling and releasing an order is used for company checks and any preparation needed for the processing of the order.
- Once an order has been released, it is ready for execution, and we can do at this time.
 - ❖ Print shop floor documents.
 - ❖ Execute goods movements.
 - ❖ Accept confirmations against the order.

Manufacturing Execution Process - Shop Floor Documents

- Shop Floor Documents are printed upon release of the Production Order, examples would be:
 - ❖ Operation-based Lists.
 - Time Tickets, Confirmation Slips.
 - ❖ Component-based Lists.
 - Material Withdrawal Slips, Pull List (consumption list).
 - ❖ PRT (Production resource/tools) Lists.
 - Overview of PRT's used with their respective operations.
 - ❖ Multi-Purpose Lists.
 - Operation Control Ticket, Object Overview.

Manufacturing Execution Process – Material Withdrawal

- When a production order is created it references a BOM to determine the necessary components to produce the material.
- It then places a reservation on each of the components.
- Upon release of the order (or operation) you can withdraw the reserved materials from inventory.
 - ❖ Reservation is updated.
 - ❖ Inventory is updated.
 - ❖ Costs are assigned to the order as actual costs.

Manufacturing Execution Process - Confirmations

- Confirmations are used to monitor and track the progression of an order through its production cycle
 - ❖ Confirmation can be done at the operation or order level.
- Exact confirmation shortly after completion of an operation is essential for realistic production planning and control.
- Data that needs confirmation include
 - ❖ Quantities: yield, scrap, rework.
 - ❖ Activity data: setup time, machine time.
 - ❖ Dates: setup, processing, teardown started or finished.
 - ❖ Personnel data: employee who carried out the operation, number of employees involved in the operation.
 - ❖ Work center.
 - ❖ Goods movements: planned and unplanned.
 - ❖ Variance reasons.
 - ❖ PRT usage.

Manufacturing Execution Process – Goods Receipt

- Acceptance of the confirmed quantity of output from the production order into stock.
 - ❖ Effects of the Goods Receipt.
 - Updates stock quantity.
 - Updates stock value.
 - Changes price stored for future valuation.
 - Updates production order.
 - ❖ Three documents are created.
 - Material document.
 - Accounting document.
 - Controlling document.

Manufacturing Execution Process – Order Settlement

- Consists of settling the actual costs incurred in the order to one or more receiver cost objects.
 - ❖ Receivers could include: a material, a cost center, an internal order, a sales order, a project, a network, a fixed asset.
- Parameters for Order Settlement.
 - ❖ Settlement Profile.
 - Specifies the receivers, distributions rules and method.
 - ❖ Settlement Structure.
 - Determines how the debit cost elements are assigned to the settlement cost elements.

Manufacturing Execution Process – Order Settlement (Cont.)

- Settlement Rule.
 - ❖ Automatically assigned on creation of order, the parameters are used to define this rule.
 - Has one or more distribution rule assigned to it.
 - Distribution rules define: cost receiver, settlement share, settlement type.

Manufacturing Execution Process – Order Settlement (Cont.)

- Settling a Production Order to Stock.
 - ❖ Debit posting is made to the Production Order with the value of the material *
 - ❖ Difference between the debt posting and credit posting is posted to a price difference account.

Material	Prod. Order	Price Diff.
80	100	20

* Material Price is determined by the quantity produced times the Standard Price in the Material Master.

Manufacturing Execution Process – Order Settlement (Cont.)

- Costs analyzed.
 - ❖ Primary.
 - Materials.
 - External Processing.
 - ❖ Secondary.
 - Production, Material, and Administrative Overhead
 - Labor.
- Cost Analysis Reporting.
 - ❖ Calculate and analyze planned costs, target costs, and actual costs of the production order.
 - ❖ Calculate and analyze variances.

In conclusion

- Manufacturing Execution Process
 - ❖ Production proposal (planning/other).
 - ❖ Capacity planning.
 - ❖ Schedule & release.
 - ❖ Shop floor documents.
 - ❖ Goods issue.
 - ❖ Completion confirmation.
 - ❖ Good receipt.
 - ❖ Order settlement.

Understand



7. Production Planning and Execution in SAP S/4HANA (using Global Bike)

**THANK YOU
FOR YOUR ATTENTION**

Q&A